

Weld County Department of Public Health and Environment

Division of Public Health Services · Community Health · Environmental Health

Mobile Health Unit Program · Greeley, Colorado

Advancing health equity through mobile community-based services across Weld County

Mobile Health Unit Data Systems Assessment

Phase 1: Data Quality Assessment

FCSM Framework for Data Quality (2020), 11 Dimensions

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All statistical values in this report were verified directly against the SAS output (MHU_Phase1_Output.pdf, 148 pages, N = 85 events, N = 315 survey responses). All figures are precision-cropped from the original SAS-generated plots. No values were estimated or generated outside of SAS.

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1. Executive Summary

This report presents the Phase 1 data quality assessment of the Mobile Health Unit programme at Weld County Department of Public Health and Environment. The assessment applies the Federal Committee on Statistical Methodology Framework for Data Quality (2020) across all 11 dimensions. Evidence draws from four structured interviews, a PHS follow-up interview, written responses from Environmental Health, both original tracking sheet workbooks, all three Read Me tabs, and a 315-response customer experience survey, combined with SAS statistical analysis of the tracking sheet dataset ($N = 85$ events). This is a mixed-methods assessment: qualitative findings from interviews and document review are presented alongside the quantitative SAS evidence that confirms, quantifies, or in some cases corrects the initial qualitative impression.

***Public Health Services:** “ If we want to go and try to get additional grant funding, the story of the data and the demographics and the services and the locations and the people we are serving, we have to be able to tell all of it to multiple audiences. That is what has to be achieved. Right now we do not have probably the tools or the system in place. ”*

The programme entered its second year in reasonable shape. Staff across all three divisions know what they are recording and why. Written definitions for all key variables were established before operations started, and every interview group confirmed these definitions independently. The data system does not need to be rebuilt; it needs targeted corrections before the Phase 3 dashboard is constructed.

The most significant structural problem is a contradiction inside the same workbook. The main Read Me tab instructs Environmental Health to record total hours including travel time. The PHS and CH tabs instruct those divisions to record onsite time only. Every division followed the instruction written for it. The result is that the staff hours column means different things depending on which division entered the data. This is not a staff error; it is a documentation conflict that must be resolved by programme leadership before any cross-division staffing analysis is presented.

Three additional findings carry urgent weight. First, address completeness collapsed from 68.5 percent in 2025 to 22.6 percent in 2026, a decline confirmed as highly significant ($\chi^2 = 14.86$, $p < 0.001$). Logistic regression identifies year alone as the driver, independent of region. Second, the tracking sheet contains no demographic data of any kind. The programme can count how many people it served but cannot describe who they were, which limits equity reporting and grant justification. Third, the master tracking sheet is stored in a personal OneDrive account, and ownership has already changed hands once due to staff turnover. This risk already materialised once under the prior owner: the original file disappeared when that staff member left the county, and only a locally saved backup prevented permanent loss.

The most promising discovery came from the Environmental Health interview. EpiTrax holds approximately 150 data points per child for every blood lead test conducted in Weld County. EH staff offered to pull all records from MHU events. Requesting this data pull is the first priority after this report is submitted.

2. Methodology

2.1 Framework

The FCSM Framework for Data Quality (2020) organises assessment across 11 dimensions in three domains: Utility (Relevance, Accessibility, Timeliness, Punctuality, Granularity), Objectivity (Accuracy and Reliability, Coherence), and Integrity (Scientific Integrity, Credibility, Computer Security, Confidentiality). Parker et al. (2024) confirmed the framework applies to blended health programme data drawn from multiple sources, which matches the structure of the MHU tracking sheet.

2.2 Design

The decision to assess data quality before building any dashboard was a direct scope change made by the project supervisor early in the engagement, not the original plan.

Public Health Services, project inception meeting, lightly edited for clarity: “We can build dashboards, but if it’s garbage in, garbage out, it wouldn’t be useful. We’d have to come back and do this all over again.”

This assessment uses a convergent mixed-methods design (Creswell and Plano Clark, 2018): qualitative and quantitative strands are collected and analysed in parallel, then merged at the interpretation stage so that each strand can confirm, challenge, or extend the other. A convergent design was chosen over a sequential explanatory or sequential exploratory alternative because the two evidence streams were available simultaneously rather than one motivating the other, and because the central analytic task here is triangulation: testing whether what staff describe in interviews matches what the tracking sheet actually contains, not using one strand to build an instrument for the other. The qualitative strand comprises four structured division interviews, one PHS follow-up interview, written responses from Environmental Health, and a full read of both tracking sheet workbooks including all three Read Me tabs. The quantitative strand comprises SAS analysis of the $N = 85$ event-level tracking sheet records and the $N = 315$ customer experience survey.

Two statistical facts govern every quantitative comparison in this report and are stated once here rather than repeated at each instance: the tracking sheet records aggregate event-level counts rather than individual patient records, and the 2025 observation window covers eight months while the 2026 window covers five months, so direct raw-total comparisons are invalid without annualisation.

2.3 How the Two Strands Are Integrated

Three integration techniques are used throughout this report, following established mixed-methods practice (Fetters, Curry, and Creswell, 2013). First, **joint displays** pair a quantitative result directly against the qualitative evidence that explains it, in a single table, so the reader sees both strands at the point of interpretation rather than in separate sections. Five joint displays appear in this report, at the points where the two strands had the most to say to each other. Second, every major finding is explicitly labelled as one of three integration outcomes: **convergence** (the two strands independently support the same

conclusion), **divergence** (the two strands appear to disagree, with the disagreement itself analysed rather than smoothed over), or **expansion** (one strand reveals something the other strand could not have detected on its own). Third, interview quotations are placed adjacent to the statistical result they speak to, rather than collected into a separate qualitative section, so that mechanism and magnitude are read together.

Table 1: Integration Outcomes Used in This Report

Outcome	Definition and an example from this report
Convergence	Qualitative and quantitative evidence independently point to the same conclusion. Example: staff describe the six-test STI standard in interview; SAS confirms median and mode both equal 6.000 across all 32 STI events.
Divergence	The two strands appear to conflict, and the conflict is itself informative. Example: the Read Me tabs give two contradictory written instructions for staff hours; SAS confirms PHS, CH, and EH staff hours ratios cannot be meaningfully compared as a result.
Expansion	One strand surfaces a finding invisible to the other. Example: the EH interview revealed the EpiTrax demographic pathway, a finding no amount of tracking sheet analysis could have produced, since EpiTrax data does not appear in the tracking sheet at all.

2.4 Limitations

Four limitations apply to this assessment and are stated here so the findings above are read with appropriate caution rather than overconfidence.

Interview sample size and representativeness. Four structured division interviews plus one PHS follow-up interview were conducted, with EH represented through written responses rather than a live interview. This is a small qualitative sample even by the standards of a single-programme assessment, and the people interviewed were primarily programme leads and senior clinical staff rather than the full range of frontline staff who enter data day to day. Findings attributed to divisions in this report (for example, “PHS practice” or “CH practice”) necessarily reflect the perspective of the specific individuals interviewed and may not capture variation across all staff within a division.

Single-assessor design. One person conducted both the qualitative interviews and the quantitative SAS analysis. This has the advantage of close familiarity between the two strands, which is part of why the integration in this report is as tight as it is, but it also removes the inter-rater reliability check that a multi-assessor qualitative analysis would normally provide. No second coder independently reviewed interview transcripts for theme identification.

Social desirability risk. Staff were interviewed by someone reporting findings to their own supervisor and to programme leadership. Several interview responses describe practices in a self-justifying frame (for example, explanations of why an apparent anomaly is not actually an error). Where possible this report cross-references such explanations against the tracking

sheet data itself, as in the Centennial Library and STI ratio examples in Section 5.9, but explanations that could not be independently verified are flagged as such rather than accepted at face value.

Small quantitative subsamples. Several quantitative comparisons in this report, particularly for Environmental Health, draw on samples of fewer than 20 observations. The formal power analysis in Section 5.5 addresses this directly for EH; the same caution applies, to a lesser degree, to several CH 2026 comparisons where $n = 13$. Where quantitative power is limited, qualitative evidence is given comparatively more interpretive weight, and this is noted explicitly at each such point rather than left implicit.

A quantitative finding stronger than its qualitative explanation. Address completeness fell from 68.5% to 22.6% between 2025 and 2026, confirmed highly significant by four independent statistical tests and isolated to year rather than region by logistic regression. A working qualitative explanation exists from the project supervisor (a staff education and prioritisation issue rather than a system failure), and address quality was written into every division’s interview guide as a priority question, but no division interview had confirmed or ruled out that explanation directly with frontline staff at the time of this report. This is flagged in Section 5.7.2 as the clearest instance in the report where the quantitative evidence is more solid than the qualitative evidence behind it, and is named here as a limitation of how far this assessment’s qualitative strand reached, not as a gap that was investigated and came up empty.

2.5 Rating Scale

Table 2: FCSM Dimension Rating Scale

Rating	Meaning
HIGH	Quality is strong, well-documented, and threats are minimal or actively managed.
MEDIUM-HIGH	Generally strong with minor addressable gaps.
MEDIUM	Functional with documented inconsistencies or gaps requiring attention.
LOW-MEDIUM	Notable threats present; reliability is limited in meaningful ways.
LOW	Significant gaps; data is not reliably usable without remediation.

3. Variable Definitions: What the Read Me Says versus What Divisions Do

Before FCSM ratings can be assigned with confidence, the question of what each variable actually measures in practice has to be settled. The Read Me tabs provide three separate sets of guidance that do not always agree.

Before the structured division interviews were conducted, Public Health Services explained the engagement variable during a project scoping meeting. That explanation is important

because it confirms how PHS applies the definition in day-to-day practice, ahead of any formal interview prompting.

Public Health Services, project scoping meeting: “ *I can say when, I can say where, I can say what, I can say how many. I cannot very well say who.* ”

The table below maps every key variable to its written guidance, its observed practice in each division, and whether that practice is consistent with the written standard.

Table 3: Key Variable Definitions: Written Standard versus Observed Practice

Variable	Division	Written Standard	Assessment Finding
Engagement	PHS/CH	Read Me (PHS and CH tabs): number of interactions with individuals	CH practice is closest. PHS and EH count individuals instead.
Engagement	EH	No EH-specific Read Me tab exists	No written definition; EH departs from the agreed standard.
Staff Hours	PHS/CH	Onsite time only; travel and vehicle checks excluded	Written standard exists; cross-division comparison needs verification.
Staff Hours	EH	Main Read Me: total hours from leaving office to returning, travel included	CONFLICT: two contradictory written instructions in the same workbook.
People Served	PHS	No Read Me definition	Verbally confirmed: direct recipients entered into Athena only.
Health Screenings	CH	Read Me lists A1C, BMI, lipids, blood pressure (four tests)	Read Me is outdated; current practice uses eight tests.
OOR Results	CH	CH Read Me: count of out-of-range or positive test results	Resolved: counts individual test results, not individuals.
STI Tests	PHS	Read Me lists column names; no counting guidance given	Gap: six-test standard confirmed in interview but not in writing.
Blank Cells	All	If a service was not offered, leave blank; do not enter zeros	Confirmed in all Read Me tabs. Critical for interpreting missingness.

4. FCSM Data Quality Scorecard

Table 4: FCSM Phase 1 Scorecard – MHU Programme, June 2026

Domain	Dimension	PHS	CH	EH
Utility	Relevance	HIGH	HIGH	MEDIUM
Utility	Accessibility	MEDIUM	MEDIUM	MEDIUM
Utility	Timeliness	MEDIUM	MEDIUM	LOW-MEDIUM
Utility	Punctuality	MEDIUM	MEDIUM	MEDIUM
Utility	Granularity	LOW	LOW	LOW
Objectivity	Accuracy and Reliability	MEDIUM-HIGH	HIGH	MEDIUM
Objectivity	Coherence	MEDIUM	MEDIUM	MEDIUM
Integrity	Scientific Integrity	MEDIUM-HIGH	MEDIUM-HIGH	MEDIUM-HIGH
Integrity	Credibility	HIGH	HIGH	HIGH
Integrity	Computer Security	MEDIUM	MEDIUM	MEDIUM
Integrity	Confidentiality	HIGH	HIGH	HIGH

5. FCSM Dimension Findings

5.1 5.1 Relevance – PHS: HIGH | CH: HIGH | EH: MEDIUM

Relevance is whether the data collected serves the programme’s actual needs.

PHS runs the most data-intensive operation on the mobile unit. Every clinical event feeds four systems simultaneously: STI grant reporting goes to CDPHE through REDCap, immunisation records sync automatically to the Colorado Immunisation Information System through Athena, patient care is documented in Athena, and programme-level data goes into the MHU tracking sheet. That is a significant data management burden distributed across a small clinical team, and it directly explains several timeliness and coherence findings discussed later in this report.

***Public Health Services:** “ Public health services is probably the most data heavy of all the groups. ”*

The STI testing ratio is a specific example of how PHS translates a clinical protocol into a data standard. The programme offers a fixed panel of six tests, but not every person receives all six, and the ratio between tests administered and people tested reflects exactly that variation.

***Public Health Services:** “ We routinely offer six STI tests. Not every person gets all of them – it depends on risk assessment and consent. ”*

On the Narcan and harm reduction services that appear only sporadically in the notes field, the recording standard was explicit even if it was never written down in the Read Me.

Public Health Services: “ *If I am not providing the Narcan education I am not counting it as a service that we are providing that day.* ”

Community Health frames its engagement variable specifically because it captures outreach conversations and referrals that would otherwise disappear entirely if CH only counted formal screenings.

Community Health: “ *If we are only counting the screenings we are not counting a big part of what we are doing which is the outreach and making connections to other resources.* ”

Environmental Health is rated Medium for Relevance not because of poor data quality but because EH’s core mission is largely regulatory rather than clinical, and blood lead testing is the one service that genuinely fits the mobile unit model. EH staff described a deliberate targeting decision about where they bring lead testing equipment.

Environmental Health: “ *We have analysed our impact at 2025 events and decided to only bring blood lead testing to events focused on health screenings where children are expected.* ”

This quote is treated in this report as evidence for a deliberate *targeting* decision: EH chose to bring testing equipment only to events where children were expected. It is treated with more caution as evidence for the much larger claim, made elsewhere in this report, that the 78% drop in EH engagements and the near-total 2026 absence (one event attended in five months) was entirely strategic. The Fisher exact test on EH attendance by event type (SAS PROC FREQ, page 134, $p = 0.837$) only establishes that attendance is not explained by whether an event was public or private; it cannot independently confirm that the scale of the 2026 decline was a deliberate choice rather than partly reflecting reduced EH staff capacity or competing regulatory workload, neither of which was asked about directly in the EH interview. This is flagged as an unresolved tension rather than smoothed into a single confident narrative: the qualitative evidence supports *some* intentional narrowing, and the quantitative evidence is silent on whether that narrowing fully accounts for the magnitude observed.

With only one event attended in the first five months of 2026, the EH dataset is too small for statistical conclusions, a constraint formally confirmed by the power analysis in Section 5.5.

5.1.1 Why Non-Parametric Tests Are Used Throughout This Report

Standard tests such as the t-test assume data follows a bell curve. When that assumption fails, the t-test produces unreliable results. The Shapiro-Wilk W statistic tests this assumption: W close to 1.0 means bell-shaped; W far below 1.0 means the assumption fails. A p-value below 0.05 confirms the failure and requires non-parametric alternatives. The Mann-Whitney test ranks all values together and asks whether one group consistently receives higher ranks; it works correctly regardless of distribution shape.

Table 5: Normality Test Results (SAS PROC UNIVARIATE, pages 2–32). All four tests (Shapiro-Wilk, Kolmogorov-Smirnov, Cramer-von Mises, Anderson-Darling) agree.

Variable	2025 W	2025 p	2026 W	2026 p	Verdict
STI Ratio	0.741022	0.0007	0.755083	0.0005	Non-normal both years
PHS Engagement Ratio	0.440516	<0.0001	0.818131	0.0013	Non-normal both years
CH Screening Ratio	0.715091	<0.0001	0.960220	0.7571	Non-normal 2025 only
CH OOR Rate	0.815269	0.0002	0.944617	0.5194	Non-normal 2025 only

Interpretation: Every key ratio variable fails the normality test in 2025, and three of the four also fail in 2026. The Shapiro-Wilk W values range from 0.441 to 0.815 against a maximum of 1.0, and all associated p-values are below 0.001, well beneath the 0.05 threshold. This means the bell-curve assumption required by standard parametric tests such as the t-test and ANOVA does not hold for this dataset. Applying those tests anyway would produce incorrect p-values and confidence intervals. The Mann-Whitney test (for two-group comparisons) and the Kruskal-Wallis test (for comparisons across three or more groups) make no distributional assumptions: they work on ranks rather than raw values and remain valid regardless of how skewed or heavy-tailed the data are. These tests are used for every year-on-year and region-by-region comparison in this report. The normality tests are therefore not a negative finding but a necessary diagnostic step that confirms the appropriateness of the methods used throughout.

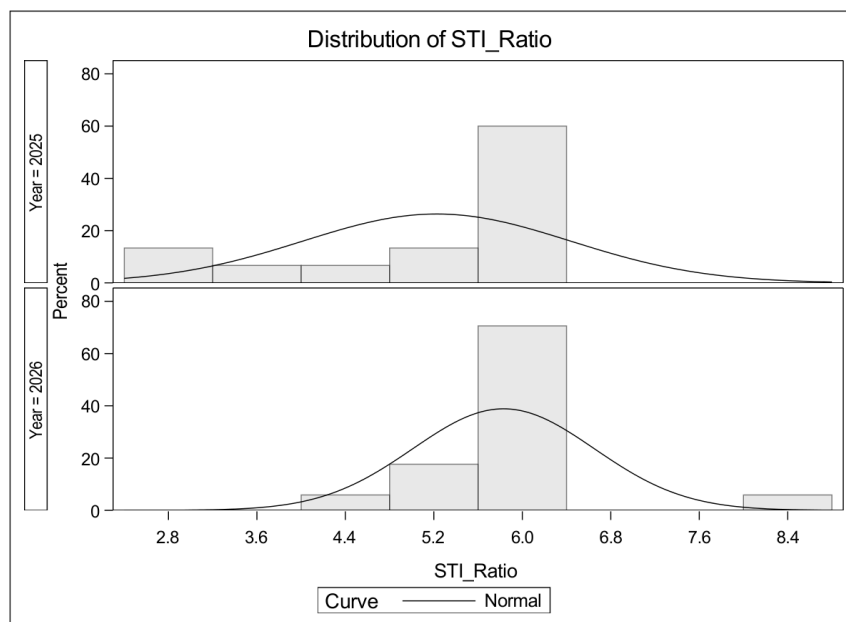


Figure 1: STI Ratio distribution by year (SAS PROC UNIVARIATE, page 6). Each bar shows how often a given ratio value occurred across events. The smooth curve is the fitted normal (bell) distribution for comparison. Values pile up at exactly 6.0 rather than spreading like a bell, reflecting the six-test standard that PHS follows. This confirms non-normality ($W = 0.741$, $p = 0.0007$ in 2025; $W = 0.755$, $p = 0.0005$ in 2026).

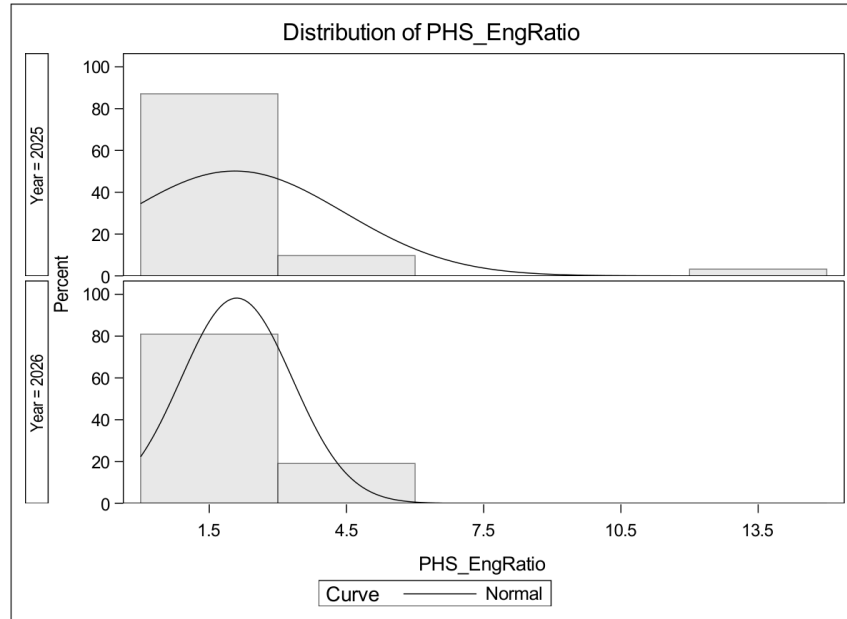


Figure 2: PHS Engagement Ratio distribution by year (SAS PROC UNIVARIATE, page 14). The single bar far to the right in the 2025 panel (top) is one event where PHS recorded 14 engagement interactions per person served. That outlier drives the very low $W = 0.441$, skewness of 4.50, and kurtosis of 22.33 – all extreme values. The 2026 panel (bottom) is compact, consistent with improved definition application.

5.1.2 Missing Data Profile

Table 6: Missing Data Summary (SAS PROC FREQ, pages 33–39; SAS PROC MEANS, page 33). Continuity-adjusted chi-square tests whether missingness is associated with year.

Variable	N Present	N Missing	χ^2 (cont. adj.)	p -value	Pattern
PHS_Engage	57	28	0.117	0.733	Structural
STI_Tests	35	50	4.701	0.030	Systematic*
CH_Engage	42	43	0.671	0.413	Structural
EH_Engage	11	74	2.843	0.092	Structural†
BLT_Count	7	78	–	–	Structural
Addr_Complete	85	0	–	–	No missing

* More STI events occurred in 2026 proportionally – programme growth, not a recording error.

† Fisher exact $p = 0.050$, borderline; likely driven in part by EH strategic withdrawal; see Section 5.1 for the unresolved scope of that explanation.

Interpretation: The continuity-adjusted chi-square test is the appropriate statistic for two-by-two tables of this size, as it corrects for continuity in the normal approximation. A p -value above 0.05 means the observed imbalance in missing data between years could easily have arisen by chance, which tells us there is no systematic relationship between year

and missingness for PHS_Engage ($p = 0.733$), CH_Engage ($p = 0.413$), and EH_Engage ($p = 0.092$). This is precisely what we expect given the Read Me instruction that blanks mean the service was not offered: a service like blood lead testing or CH health screenings is simply not delivered at every event, and those absences are distributed across both years without a systematic pattern. The one exception is STI_Tests ($p = 0.030$), which is technically systematic but for the right reason: proportionally more STI events occurred in 2026 (18 of 31 total events, 58.1%) than in 2025 (17 of 54 events, 31.5%). This reflects deliberate programme expansion of STI outreach, not a data recording inconsistency. The Address_Complete variable shows zero missing values across all 85 events, which means the completion status was recorded for every event in the dataset – only the address content itself is sometimes absent. BLT_Count has 78 missing values because blood lead testing was only offered at 7 of 85 events, a structural pattern driven by EH’s targeted deployment strategy.

5.1.3 Service Volumes by Year

Table 7: PHS Service Volumes (SAS PROC MEANS, page 44). Dataset $N = 85$ total events.

Variable	n	Sum	Mean	Med.	SD	n	Sum	Mean	Med.	SD
2025						2026				
PHS Engagements	35	354	10.11	8.00	9.11	22	457	20.77	15.00	20.50
People Served	35	248	7.09	5.00	8.99	22	292	13.27	8.00	13.26
STI Tests	17	475	27.94	24.00	25.94	18	874	48.56	35.00	46.85
Imm. Count	25	203	8.12	4.00	13.70	14	124	8.86	4.00	10.59

Table 8: CH and EH Service Volumes (SAS PROC MEANS, pages 45–46). For EH, SAS PROC MEANS (page 46) only reported N , Sum, and Mean; median and SD were not computed. Symbol – denotes not reported in SAS output. All EH figures are descriptive only; see power analysis in Section 5.5.

Variable	n	Sum	Mean	Med.	SD	n	Sum	Mean	Med.	SD
2025						2026				
CH Engagements	29	692	23.86	14.00	25.44	13	439	33.77	31.00	19.58
CH Screen Count	29	2081	71.76	33.00	92.70	13	1109	85.31	52.00	88.65
CH OOR Results	29	565	19.48	7.00	27.66	13	331	25.46	22.00	25.05
EH Engagements	10	237	23.70	n/a*	n/a*	1	32	32.00	n/a [†]	n/a [†]
Blood Lead Tests	6	16	2.67	n/a*	n/a*	1	4	4.00	n/a [†]	n/a [†]

5.1.4 Mann-Whitney Tests: Did Per-Event Service Volumes Change?

The Mann-Whitney test ranks all values from both years together and asks whether 2026 events consistently receive higher ranks than 2025 events. The Wilcoxon statistic (W) is

the sum of 2026 ranks. Z is the number of standard deviations the result sits from the null hypothesis of no difference. Effect size r follows Cohen’s benchmarks: 0.1 = small, 0.3 = medium, 0.5 = large.

Table 9: Mann-Whitney Test: PHS Engagements per event.

Statistic	Value (from SAS)	What it tells us
2025: $n = 35$, median	8.0 engagements/event	IQR = [4.5, 13.0]; mean = 10.11
2026: $n = 22$, median	15.0 engagements/event	IQR = [10.0, 20.0]; mean = 20.77
Wilcoxon W	832.0	Sum of 2026 ranks across all events
Z score	3.1793	3.18 standard deviations from no difference
p (two-sided)	0.0015	Significant – well below the 0.05 threshold
Effect size r	0.504	Large (Cohen’s benchmark: large ≥ 0.5)
95% CI on r	[0.265, 0.685]	Entire interval lies above 0.25

PHS Engagements Per Event (SAS PROC NPAR1WAY, page 50) **Interpretation:** The median number of PHS engagement interactions per event nearly doubled, rising from 8.0 in 2025 to 15.0 in 2026. Looking at the Wilcoxon score distribution plot, the entire 2026 box sits above the 2025 box: the bottom of the 2026 box (25th percentile) is roughly level with the median of 2025, meaning the lowest-ranked 2026 events performed as well as or better than the typical 2025 event. The Z score of 3.18 means the observed separation between the two years is 3.18 standard deviations away from what we would expect under the null hypothesis of no difference, a very large distance from chance. The p -value of 0.0015 means this result would arise by chance fewer than 2 times in every 1,000 repetitions of the study. The effect size $r = 0.504$ is classified as large by Cohen’s conventions, and the 95% confidence interval [0.265, 0.685] lies entirely above 0.25, meaning that even the lower bound of the plausible range represents a medium-to-large effect. This is the strongest and most clearly confirmed per-event finding in the relevance analysis. PHS is engaging meaningfully more people per event in 2026, and the evidence for this conclusion is robust across every statistical measure examined.

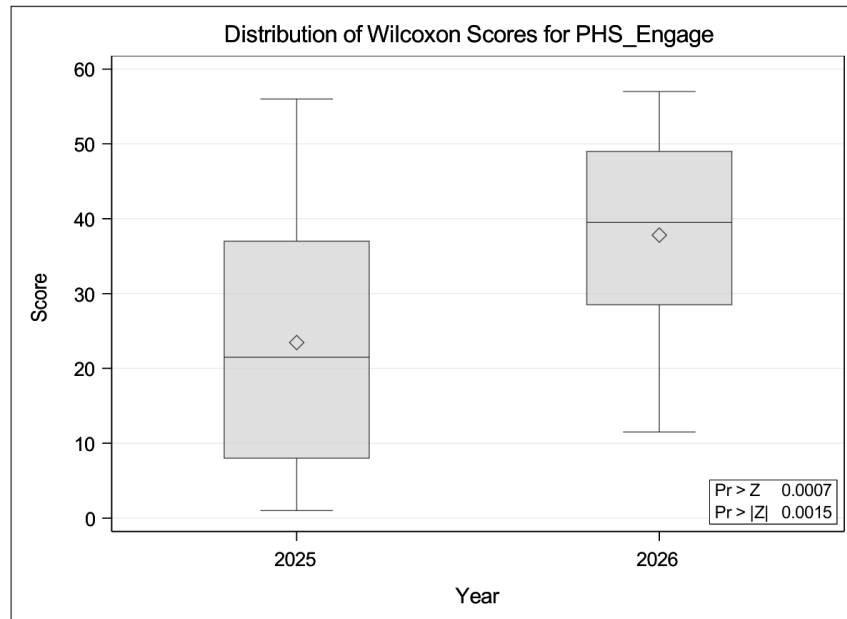


Figure 3: Wilcoxon score distribution: PHS Engagements per event (SAS PROC NPAR1WAY, page 51). Each box spans the middle 50% of rank scores for that year. The horizontal line inside is the median rank; the diamond is the mean rank. The 2026 box sits clearly above the 2025 box, confirming that 2026 events received consistently higher ranks. $\text{Pr} > |Z| = 0.0015$ in the plot corner confirms statistical significance.

Table 10: Mann-Whitney Test: PHS People Served per event.

Statistic	Value (from SAS)	What it tells us
2025: $n = 35$, median	5.0 people/event	mean = 7.09
2026: $n = 22$, median	8.0 people/event	mean = 13.27; 60% higher median
Wilcoxon W	761.5	
Z score	2.0214	
p (two-sided)	0.0432	Significant – below 0.05
Effect size r	0.321	Medium
95% CI on r	[0.059, 0.541]	Lower bound positive; direction confirmed

PHS People Served Per Event (SAS PROC NPAR1WAY, page 52) Interpretation: The median number of distinct people served per event rose from 5.0 in 2025 to 8.0 in 2026, a 60% increase at the median level. The mean also rose substantially, from 7.09 to 13.27 per event. In the Wilcoxon score distribution plot the 2026 box sits above the 2025 box but with clear overlap, which is the visual signature of a medium rather than large effect – the two years are meaningfully different but not as cleanly separated as the engagement comparison. The Z score of 2.02 and p -value of 0.043 confirm the result crosses the conventional significance threshold, though with less margin than the engagement finding

($p = 0.0015$). The effect size $r = 0.321$ falls squarely in the medium range. The confidence interval $[0.059, 0.541]$ has a positive lower bound, meaning even in the most conservative scenario the true effect is in the upward direction. Taken together with the engagement finding, these two results paint a consistent picture: in 2026 PHS events were simultaneously serving more individual people and delivering more engagement interactions per person, which points to both broader reach and greater service depth per event.

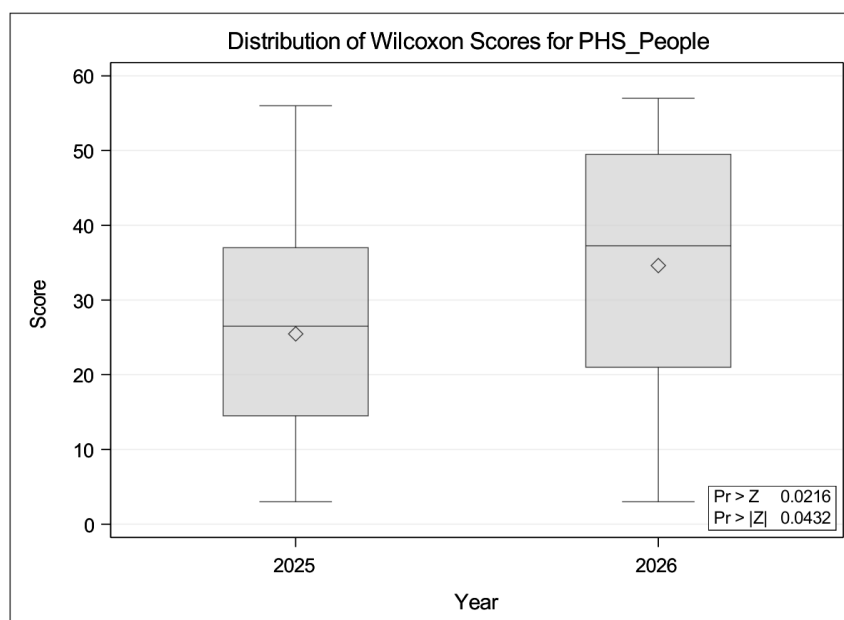


Figure 4: Wilcoxon score distribution: PHS People Served per event (SAS PROC NPAR1WAY, page 53). The 2026 box sits above the 2025 box. The boxes overlap somewhat, consistent with a medium rather than large effect size. $\text{Pr} > |Z| = 0.0432$ confirms statistical significance.

Table 11: Mann-Whitney Test: CH Engagements per event. Annualised totals from SAS PROC PRINT, page 58.

Statistic	Value (from SAS)	What it tells us
2025: $n = 29$, median	14.0 engagements/event	mean = 23.86
2026: $n = 13$, median	31.0 engagements/event	mean = 33.77; per-event rate rising
Wilcoxon W	344.5	
Z score	1.7567	
p (two-sided)	0.0790	Borderline; just above 0.05
Effect size r	0.345	Medium
95% CI on r	[0.037, 0.593]	Entirely above zero
Raw total 2025	692 engagements	8 months of data
Raw total 2026	439 engagements	5 months of data; apparent decline only
Annualised 2025	1,038/year	$692 \div 8 \times 12$
Annualised 2026	1,054/year	$439 \div 5 \times 12$; essentially stable

CH Engagements Per Event (SAS PROC NPAR1WAY, page 54) **Interpretation:** The raw total of 692 in 2025 versus 439 in 2026 looks like a 37% decline, and this framing would be incorrect and misleading. The 2025 dataset covers eight calendar months (May through December) while the 2026 dataset covers only five months (January through May). Comparing these raw totals directly is equivalent to comparing a runner's cumulative distance at mile 8 of one race to mile 5 of another and declaring the runner to have slowed down. The comparison is structurally invalid. When both figures are annualised to the same 12-month basis – dividing by months observed and multiplying by 12 – the two years produce almost identical results: 1,038 and 1,054 engagements per year, a difference of less than 2%. At the per-event level, CH actually delivered substantially more in 2026: the median per-event count rose from 14.0 to 31.0, and the mean from 23.86 to 33.77. In the Wilcoxon score distribution plot the 2026 box sits clearly above the 2025 box despite the lower raw total, confirming the per-event rate is increasing. The p-value of 0.079 is just above the 0.05 threshold, making this a borderline result, but the confidence interval [0.037, 0.593] lies entirely above zero, which means we can be 95% confident the true direction of change is positive. This finding must not be presented to leadership as a programme decline. It should be presented as: the CH engagement rate per event is growing strongly, annualised totals are flat, and the apparent raw decline is a counting artefact of an unequal observation window.

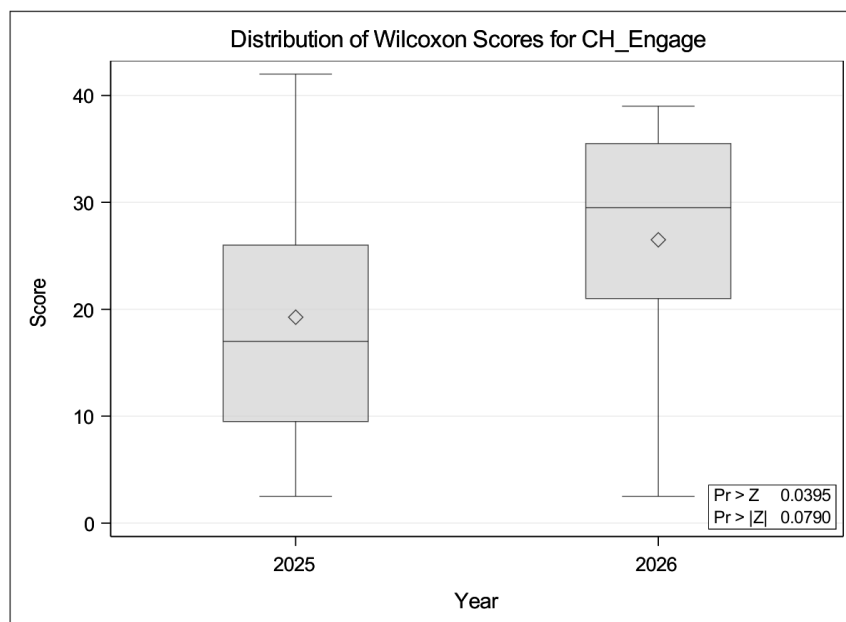


Figure 5: Wilcoxon score distribution: CH Engagements per event (SAS PROC NPAR1WAY, page 55). Despite the lower raw total, the 2026 box sits above the 2025 box (mean rank 26.5 vs 19.3), confirming that 2026 events produced more engagements per event. $\text{Pr} > |Z| = 0.079$: borderline.

Table 12: Mann-Whitney Test: STI Tests per event.

Statistic	Value (from SAS)	What it tells us
2025: $n = 17$, median	24.0 tests/event	17 STI events in 8 months
2026: $n = 18$, median	35.0 tests/event	18 STI events in 5 months
Wilcoxon W	263.5	
Z score	-1.3884	
p (two-sided)	0.1650	Not significant
Effect size r	0.259	Small
Annualised 2025	713 tests/year	$475 \div 8 \times 12$
Annualised 2026	2,098 tests/year	$874 \div 5 \times 12$; nearly tripled

STI Tests Per Event (SAS PROC NPAR1WAY, page 56) **Interpretation:** The Mann-Whitney test on per-event STI test counts is not significant ($Z = -1.388$, $p = 0.165$), and the effect size $r = 0.259$ is small. Looking at the Wilcoxon score distribution plot, the two boxes overlap substantially with no clear separation – this is the expected visual appearance when no meaningful per-event difference exists. The 2025 median was 24.0 tests per event and the 2026 median was 35.0 tests per event, but the high variability in both years (SD of 25.94 and 46.85 respectively) means this difference does not reach statistical significance. However, the programme-level picture is very different. STI test delivery nearly tripled on an annualised basis, from 713 tests per year in 2025 to 2,098 per year in 2026. This growth came not from making individual events larger but from dramatically increasing the frequency of

STI events: 17 STI events in eight months of 2025 versus 18 STI events in only five months of 2026, meaning the event rate more than doubled. The STI programme is scaling through volume, not size – more events spread across more locations and dates, rather than larger single events. This is an operationally important distinction for grant reporting and capacity planning.



Figure 6: Wilcoxon score distribution: STI Tests per event (SAS PROC NPAR1WAY, page 57). The two boxes overlap substantially, consistent with no significant per-event difference ($p = 0.165$). The real STI growth comes from the increased frequency of STI events across the year.

5.1.5 Annualised Comparison (SAS PROC PRINT, page 58)

Table 13: Annualised Service Volume Comparison. 2025 = 8 months; 2026 = 5 months. Annualised = raw $\div$ months $\times$ 12.

Metric	2025 Raw	2025 Ann.	2026 Raw	2026 Ann.	Ann. Change
CH Engagements	692	1,038	439	1,054	+2% – stable
PHS Engagements	354	531	457	1,097	+107% – large
PHS People Served	248	372	292	701	+88% – large
STI Tests	475	713	874	2,098	+194%
CH Screenings	2,081	3,122	1,109	2,662	–15% – broadly stable
EH Engagements	237	356	32	77	–78% – attributed to EH withdrawal
Blood Lead Tests	16	24	4	10	–58% – attributed to EH withdrawal

The seasonal sensitivity check (SAS PROC MEANS, page 59) confirms the PHS annualised increase is genuine and not a seasonal artefact. The 2026 April mean of 31.4 engagements per event and May mean of 40.0 far exceed the 2025 June through December baseline mean of 10.2. If the annualised increase were simply a consequence of 2026 data being collected during a seasonally busier period, we would expect 2026 months to mirror similarly active 2025 months. Instead, the 2026 figures substantially exceed even the highest 2025 months, ruling out a seasonal explanation and confirming that the increase reflects genuine programme growth in PHS engagement capacity.

5.2 5.2 Accessibility – All Divisions: MEDIUM

Accessibility is the ease with which data users can obtain the data they need in a usable form. All three divisions currently receive their programme data through one product: the quarterly report produced from the master tracking sheet. The data team confirmed the production process is largely automated through Tableau, but that automation is also the only way to access the data; no division can currently run an ad hoc query without requesting analyst time. Cancelled events represent service demand the programme could not meet and are relevant to capacity and planning analysis, but they are not currently surfaced anywhere outside the raw tracking sheet.

Two barriers drive the Medium rating: divisions cannot run custom queries independently, and no confirmed publishing pathway exists for the external Tableau dashboard envisioned for Phase 3.

The future dashboard work referenced throughout this report is intended to serve two distinct audiences with different needs, a distinction the supervisor drew explicitly before any Phase 3 design work began.

Public Health Services, project inception meeting, lightly edited for clarity: “
The teams are the internal audience. The external audience is my superiors, the board, the general public. We don’t want to bog these people down with details. How many STI tests did you give at this particular event? They don’t care about that. They want to see a high-level roll-up. ”

This distinction matters for how Phase 3 should be scoped: an internal dashboard answering operational questions (which locations are worth revisiting, how staffing relates to volume) is a different design problem from an external dashboard answering accountability questions (who is being served, what is the programme’s overall reach), and conflating the two risks building something that satisfies neither audience well.

Data Team: “ *We pull the Excel file, upload it to Tableau, hit update, use the date filter. It is pretty automated. The biggest thing is we need a date range filter by month, not just by year. ”*

Community Health: “ *What is missing is the interactive component – being able to enter a time period and having it update. Being able to say I want to look at events in the Greeley region only. ”*

Survey language data provides the best available population proxy. Of 315 survey respondents, 198 (63.06%) completed the survey in English and 116 (36.94%) completed it in Spanish (SAS PROC FREQ, page 60). Event type shifted significantly toward strategic deployment: strategic events rose from 27.8% of all events in 2025 to 56.7% in 2026 ($\chi^2 = 7.692$, $df = 2$, $p = 0.021$, SAS PROC FREQ, page 62).

5.3 5.3 Timeliness – PHS: MEDIUM | CH: MEDIUM | EH: LOW-MEDIUM

Timeliness is the lag between when an event occurs and when reliable data is available. The PHS data entry workflow illustrates why this dimension carries real operational risk. After a clinical event, the nurse on duty completes charting for each patient seen, and that chart eventually has to be transcribed into the MHU tracking sheet by a programme coordinator.

Public Health Services: “ *The coordinator’s desk so she can put the data into her spreadsheet. ”*

Public Health Services: “ *When we have two nurses, everything is done in real time and when we came back all I had to do was the labs. When it is just the two of us, I chart and put everything into REDCap and Athena after we return. ”*

The REDCap transfer adds specific risk on top of this. STI data must be manually transferred from Athena to REDCap because no bidirectional sync exists between the two systems, a known issue for over two years. PHS also maintains two separate tracking sheets: a general outreach sheet and the MHU master sheet, with events transferred manually between them. Every manual transfer point is a place where a record can be delayed, mistyped, or dropped

entirely, and this manual-transfer risk is the central driver of the Medium rating for PHS and CH.

Environmental Health, which attends the fewest events of any division, takes a more relaxed but still functional approach to timeliness.

Environmental Health, written response: “ Almost always will be entered by the same staff member, with a backup. We attend the least often of all divisions so I try to enter monthly. ”

Table 14: Event Duration by Year (SAS PROC MEANS, page 66).

Year	N	Mean (hr)	Median (hr)	SD	Min	Max
2025	52	3.74	3.00	1.71	1.50	9.00
2026	30	3.22	3.00	1.23	2.00	6.00

Interpretation: The median event duration is exactly 3.0 hours in both years. The SD declined from 1.71 to 1.23, meaning events ran more consistently in 2026. The primary timeliness concern is not event duration but the manual charting workflow, particularly the un-audited transfer between the outreach tracking sheet and the MHU master sheet.

5.4 Punctuality – All Divisions: MEDIUM

Table 15: Events by Quarter (SAS PROC FREQ, pages 67–69).

Quarter	All 2025	All 2026	PHS 2025	PHS 2026	CH 2025	CH 2026
Q1 (Jan–Mar)	0	20	0	14	0	8
Q2 (Apr–Jun)	14	11	6	8	10	5
Q3 (Jul–Sep)	22	0	16	0	15	0
Q4 (Oct–Dec)	18	0	14	0	5	0
Total	54	31	36	22	30	13

Data Team: “ We as a group all sat down and determined how each of these things are counted so there is consistency. I should put a quarterly reminder in to remind people that the Read Me is there. ”

Community Health: “ You are highlighting that we probably need to have a reminder and discuss it more often. ”

Interpretation: Events occurred in every available quarter with no unexplained gaps. The Medium rating reflects a process gap: the only mechanism for ensuring divisions update the tracking sheet before the quarterly data pull is a single reminder email, with no verification

step confirming completion. Both the data team and Community Health independently acknowledged during interviews that a more regular reminder and discussion of the Read Me definitions would be useful, which is itself evidence that the gap is recognised even if not yet acted on.

5.5 5.5 Granularity – All Divisions: LOW

Granularity is the degree to which data can be disaggregated to describe subpopulations. The tracking sheet holds no individual-level or demographic data for any division. It records aggregate counts at the event level: how many people were served and what services were delivered. It cannot say who those people were.

The supervisor's own explanation for this gap, given in the project inception meeting before any division interviews took place, frames it as a structural consequence of who the tracking sheet was built for rather than an oversight. Frontline clinical staff running a mobile unit event are focused on delivering services to whoever arrives; the question of demographic composition is a different kind of question asked by a different kind of audience.

Public Health Services, project inception meeting, lightly edited for clarity:

“ If you try to tell the story of the mobile unit to the community, they’re gonna love the data about what we did. But is there the story of who we’re serving in here? No. Because we’re not capturing gender. We’re not capturing age. ”

This was not only the supervisor's view. Division leads across PHS and CH independently reached the same conclusion when asked directly.

Public Health Services: *“ I can say when, I can say where, I can say what, I can say how many. I cannot say who we are serving. ”*

Community Health: *“ I cannot directly state that we are serving priority populations. I can assume it but I cannot state it. ”*

This is not framed as a data entry failure. The tracking sheet was originally built to answer operational questions: how many events, how many services, what kind. It was not built to answer the research and accountability questions that grant reporting and equity analysis require. The practical effect is a granularity gap, and it is the subject of the formal power analysis below for EH and the four demographic pathways discussed afterward for all three divisions.

Table 16: Statistical Power Analysis: EH Blood Lead Tests. $N = 20$ total tests (SAS PROC MEANS, page 71). $\alpha = 0.05$, two-tailed.

Effect d	Meaning	Power	Practical implication
0.2	Small change	9.2%	Would miss a real effect 91% of the time
0.5	Meaningful change	35.2%	Would miss a real effect 65% of the time
0.8	Substantial change	71.6%	Would miss a real effect 28% of the time
Need $n = 63$ for 80% power at $d = 0.5$; $n = 85$ for 90% power.			
All EH inferential analysis in this report is descriptive only.			

Interpretation: Statistical power is the probability that a test detects a real effect when one genuinely exists. With only 20 blood lead tests across two years, the test would fail to detect a meaningful real difference 65% of the time. A test that fails more often than it succeeds is not useful. EH data is not bad; the sample is simply too small for formal statistical claims. Every EH number in this report is a descriptive observation only.

Four potential demographic pathways were identified across the structured interviews and a supervisory one-on-one meeting, three of them confirmed and ready to act on, the fourth pending an access decision.

Pathway 1: PHS through Athena EMR. Every PHS patient seen on the mobile unit is entered into Athena under the Mobile Unit service location. A contracted data analyst can filter records by service location and date range to produce de-identified demographic reports. Variables available include age, sex, ethnicity, insurance status, and zip code.

Pathway 2: EH through EpiTrax. This is the most significant discovery of the entire assessment. The richest individual-level data source in the MHU programme was completely invisible in the tracking sheet and emerged only through the EH interview.

***Environmental Health:** “ I have all the test results for all the children in this county from all providers. I can filter out the tests that we have done as the clinician for lead. There are about 150 data points for each individual person. I can pull it for any time frame you want. ”*

Acting on this offer is an immediate priority following this report, and is the first recommendation in Section 7.

Joint Display: *Expansion*

Table 17: Joint Display 2: The Granularity Gap and the EpiTrax Discovery

Strand	Evidence	What it establishes
Quantitative	SAS PROC CONTENTS (pp. 141–143): 0 of 49 tracking sheet variables contain demographic information.	The tracking sheet alone provides no path to demographic disaggregation; this is a closed question from the quantitative data in isolation.
Qualitative	EH interview: “There are about 150 data points for each individual person” in EpiTrax, accessible by date range and filterable to MHU events.	A demographic pathway exists, but it is invisible to any analysis of the tracking sheet, because EpiTrax is a separate state system that was never connected to the MHU dataset.

Integration outcome: expansion. The quantitative strand correctly establishes that granularity is Low and that the problem cannot be solved from within the existing dataset. Only the qualitative strand could reveal that a solution exists outside it. No amount of additional SAS analysis on the tracking sheet would have surfaced this pathway.

Pathway 3: CH through event population categorisation. CH staff can infer population characteristics from the type of event held, even without individual-level data collection.

Community Health: “Because you have to be low income to get food bank services, so we can say that.”

Pathway 4: Diabetes Prevention Programme data (pending). A fourth potential pathway emerged from the supervisory one-on-one meeting. The Diabetes Prevention Programme collects individual participant data that flows to the CDC under the programme’s federal reporting requirements. Community Health staff have indicated they cannot currently access this data directly; resolving that access question is addressed in Recommendation R1a.

5.6 5.6 Accuracy and Reliability – PHS: MEDIUM-HIGH | CH: HIGH | EH: MEDIUM

Accuracy is how closely data captures what it is supposed to measure. Reliability is whether measurement is applied consistently. The Coefficient of Variation ($CV = SD \div \text{mean} \times 100\%$) is the central consistency measure: a lower CV means the ratio took similar values across more events, suggesting the underlying definition was applied more uniformly.

Two genuine accuracy concerns surfaced during the interviews, separate from anything the quantitative analysis can detect on its own. Demographic fields in Athena are sometimes skipped during busy single-nurse events, when charting speed is prioritised over completeness. The REDCap transfer introduces a manual transcription risk because field definitions sometimes differ slightly between the two systems. The data team checks calculations and ratios for internal consistency but does not verify underlying source counts against the original

clinical record, which is an appropriate division of responsibility for their role but does mean source-level errors could in principle pass through undetected. Community Health, by contrast, has the most consistent data entry process of any division, and the three-interview evidence for this is specific rather than impressionistic.

Community Health: “ We write down the results three times: one for the client, one in the waiver room, and we have the stickers with the results from the machines. We added a fourth because sometimes we forgot to write down the A1C. If I have two staff doing the same event, each person double-checks the other’s count before we leave. ”

Public Health Services: “ I am pretty confident that we would be entering that information the same way. We circle back with one another. ”

The most important accuracy finding from the interviews concerns how the engagement variable is counted in practice. Two divisions gave specific, concrete descriptions of their counting method, and they are different from each other.

Community Health: “ If it is a family and only one person gets a service, it is just one engagement. ”

Environmental Health: “ If we speak to a family of five and we are educating that whole family, we count that as five engagements. ”

These two accounts define engagement differently: CH counts it as the number of service interactions, EH counts it as the number of people spoken to regardless of whether they received a clinical service. Both descriptions are internally coherent and reflect genuine operational practice. The quantitative CV analysis below captures the statistical consequence of this divergence without being able to identify which division is applying the written standard more closely. This is the clearest single example in the report of where the qualitative and quantitative strands complement each other: the statistics show inconsistency in how the ratio is applied; the interviews explain what the inconsistency actually consists of. The CV results below substantiate these qualitative observations quantitatively.

Environmental Health accuracy is rated Medium based on findings from the written EH responses combined with the detailed event-level tracking sheet record (SAS PROC PRINT, page 109). Two 2025 events have engagement counts that EH staff indicated were estimated retroactively from memory rather than counted in real time, which introduces a degree of recall uncertainty into a dataset that is already small. The event-level detail also resolves a specific anomaly: the Centennial Library Back to School Storytime event on 08/06/2025 recorded EH_Engage = 0 alongside BLT_Count = 12, meaning twelve blood lead tests were administered with zero recorded engagement contacts. Read against the written engagement definition (interactions, not people), this is consistent with a high-throughput testing event where every contact was a direct test administration rather than a separate outreach conversation, not a data entry omission.

5.6.1 STI Ratio: Tests Per Person Tested (Standard = 6)

Table 18: STI Ratio Summary Statistics (SAS PROC UNIVARIATE, pages 74, 76; SAS PROC MEANS, page 73).

Statistic	2025 ($n = 15$)	2026 ($n = 17$)	Note
Mean	5.2258486	5.8302356	
Median	6.0000	5.8750	
Mode	6.0000	6.0000	Confirms 6-test standard
SD	1.2108871	0.8212717	
CV	23.1711%	14.0864%	SAS PROC MEANS, p.73
95% CI on mean	[4.5553, 5.8964]	[5.4080, 6.2525]	
IQR	2.0000	0.3000	Much tighter in 2026
Skewness	-1.2441	1.6165	
Kurtosis	0.1287	6.0937	
Min / Max	2.6667 / 6.1667	4.5000 / 8.4000	

Bootstrap 95% CI on CV: 2025 = [11.1%, 28.7%]; 2026 = [5.2%, 20.2%]. The two intervals overlap, so the CV reduction from 23.2% to 14.1% is a promising directional improvement but is not yet formally confirmed at this sample size.

Interpretation: The median and mode are both exactly 6.000 in both years. In the histogram, the tallest bar in both panels sits squarely at 6.0, which is the visible signature of a fixed standard being consistently followed: most events produce a ratio of exactly 6, and the distribution is anchored around that value rather than spreading freely. The CV dropped from 23.17% in 2025 to 14.09% in 2026, a reduction of nine percentage points. Considered alone this is a promising consistency improvement, but the bootstrap confidence intervals overlap ([11.1%, 28.7%] versus [5.2%, 20.2%]), so the reduction is directional rather than formally confirmed. The IQR shrank from 2.000 to 0.300, which is a striking narrowing: the middle 50% of 2026 events all fell within 0.3 tests per person of the median, compared to a spread of 2.0 tests per person in 2025. The mean of 5.226 in 2025 and 5.830 in 2026 sit slightly below 6.0 rather than exactly at it because a small number of early events in 2025 recorded lower ratios, likely during the first months of the programme before the standard was fully embedded across all staff. The 95% confidence interval on the 2026 mean [5.408, 6.252] is entirely above 5.0 and includes 6.0, confirming the six-test standard is being met on average.

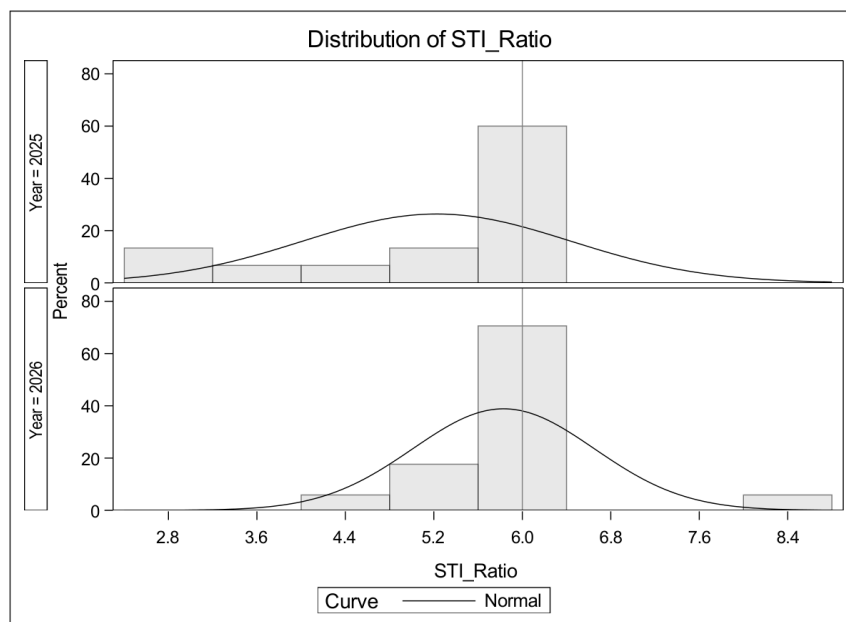


Figure 7: STI Ratio distribution by year (SAS PROC UNIVARIATE, page 78). The tall bar at 6.0 in both panels confirms the six-test standard is consistently followed. 2025 skewness = -1.244 ; 2026 skewness = 1.616 . Both deviate from the normal curve.

Table 19: Mann-Whitney Test: STI Ratio 2025 vs 2026 (SAS PROC NPAR1WAY, page 82; SAS PROC TTEST, page 84).

Test	Value (from SAS)	What it tells us
Wilcoxon W	242.0	
Z score	-0.1955	Essentially zero distance from no difference
p (two-sided)	0.8450	Not significant; far above 0.05
Effect size r	0.043	Negligible
95% CI on r	$[-0.310, 0.386]$	Crosses zero; direction not confirmed
Levene test	$p = 0.293$	No significant variance difference
F-test (Folded F)	$F = 2.17$, $df = [14, 16]$, $p = 0.1384$	No significant variance difference

Interpretation: A p -value of 0.845 and effect size of $r = 0.043$ together indicate there is essentially no difference between 2025 and 2026 in how the STI testing ratio was applied. In the Wilcoxon score distribution plot, the two boxes nearly overlap: the medians are close, the spreads are similar, and there is no visible separation between years. $Z = -0.196$ means the observed difference in rank sums is less than one fifth of a standard deviation away from zero – a negligibly small distance from the null hypothesis. The 95% confidence interval on r is $[-0.310, 0.386]$, which crosses zero in both directions, meaning the data are equally consistent with a small positive or small negative effect. Both the Levene test ($p = 0.293$) and

the F-test on variances ($F = 2.17$, $p = 0.138$) confirm that the spread of values is also not significantly different between years. When a consistency check produces a non-significant result like this – with both the central tendency and the variance stable – it is telling us the measurement process is working as intended. The STI testing standard was applied with the same consistency across both years of the programme, which is a positive credibility finding.

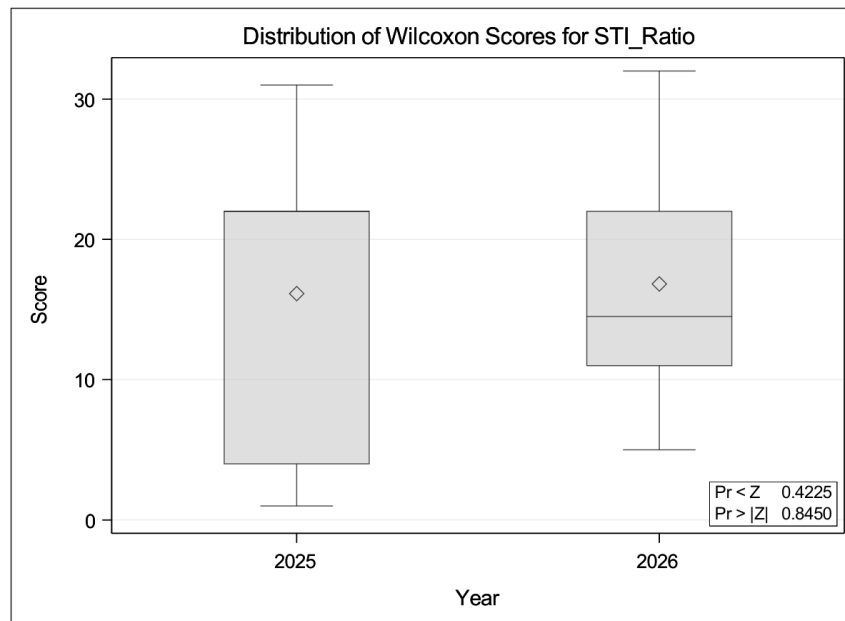


Figure 8: Wilcoxon score distribution: STI Ratio 2025 vs 2026 (SAS PROC NPAR1WAY, page 83). The two boxes nearly overlap, confirming no meaningful difference between years ($p = 0.845$).

5.6.2 PHS Engagement Ratio: Interactions Per Person Served

Table 20: PHS Engagement Ratio Summary Statistics (SAS PROC UNIVARIATE, pages 87, 89; SAS PROC MEANS, page 86).

Statistic	2025 ($n = 31$)	2026 ($n = 21$)	Note
Mean	2.0588286	2.0995647	Nearly identical
Median	1.333333	1.727273	
Mode	1.000000	2.000000	
SD	2.3862798	1.2187325	SD nearly halved
CV	115.9047%	58.0469%	SAS PROC MEANS, p.86
95% CI on mean	[1.1835, 2.9341]	[1.5448, 2.6543]	
Skewness	4.5001 (extreme)	1.4791 (moderate)	
Kurtosis	22.3273 (extreme)	1.5101 (near-normal)	
Min / Max	1.0000 / 14.0000	0.8286 / 5.0000	
Bootstrap 95% CI on CV: 2025 = [24.5%, 218.7%]; 2026 = [28.5%, 80.4%].			

CIs do not overlap: 2026 upper bound (80.4%) is below the 2025 point estimate (115.9%).

Interpretation: The mean engagement ratio is almost identical in both years (2.059 vs 2.100), which immediately tells us that the centre of the distribution did not shift. The change is entirely in the spread. In 2025, CV = 115.9% – the standard deviation exceeded the mean itself – which reflects extreme variability: one event recorded a ratio of 14.0 while many others recorded ratios of 1.0, meaning one outlier event produced 14 engagement interactions per person while the most common events produced exactly one interaction per person. This is not a plausible range if the variable is being measured consistently: a ratio of 14 is so far above any other event that it suggests either an anomalous event or a recording difference. In 2026 the maximum was 5.0, the IQR was 1.167, and CV dropped to 58.0%. Skewness fell from 4.50 (extreme positive skew, driven by the outlier) to 1.48 (moderate). Kurtosis fell from 22.33 to 1.51 – a kurtosis of 22 indicates an extremely peaked distribution with very heavy tails, which is the statistical fingerprint of a few extreme values dominating the dataset; a kurtosis near 1.5 is close to a normal distribution. The 95% CI on the 2025 mean [1.18, 2.93] is more than twice as wide as the 2026 CI [1.54, 2.65], reflecting the greater uncertainty caused by high variability. Every distributional measure – CV, skewness, kurtosis, IQR, and the width of the CI – tells the same story: the engagement variable was recorded with dramatically greater consistency in 2026 than in 2025.

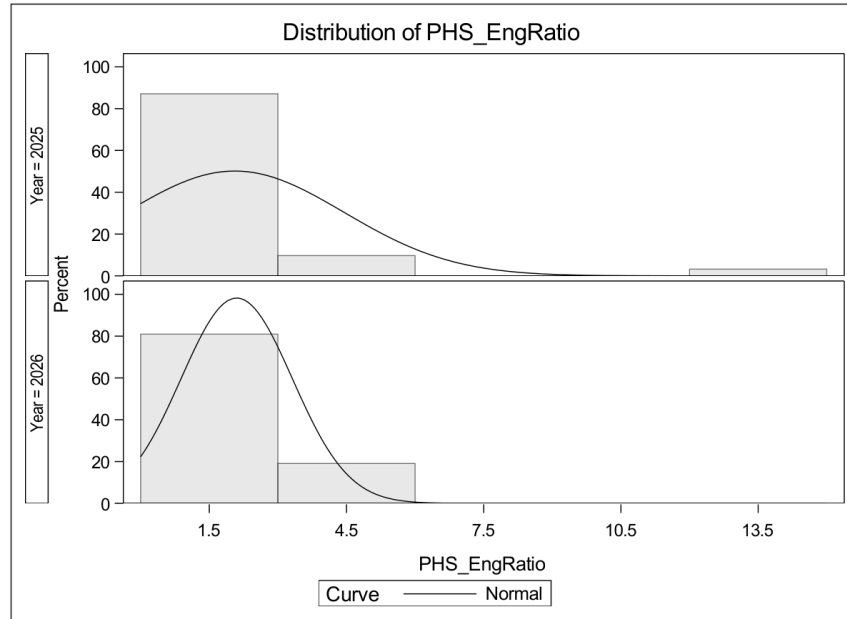


Figure 9: PHS Engagement Ratio distribution by year (SAS PROC UNIVARIATE, page 91). The single bar to the right in the 2025 panel (top) is the one event at ratio = 14.0. The 2026 panel (bottom) is compact, consistent with CV dropping from 115.9% to 58.0%.

Table 21: Mann-Whitney and Variance Tests: PHS Engagement Ratio (SAS PROC NPAR1WAY, page 95; SAS PROC TTEST, page 97).

Test	Value (from SAS)	What it tells us
Wilcoxon W	636.0	
Z score	1.4789	
p (two-sided)	0.1392	Not significant; medians similar
Effect size r	0.244	Small
95% CI on r	$[-0.035, 0.488]$	Includes zero
Levene test	$W = 0.075, p = 0.786$	Not significant
F-test (Folded F)	$F = 3.834, df = [30, 20],$ $p = 0.0027$	Variance formally reduced in 2026

Interpretation: The Mann-Whitney test ($W = 636.0, Z = 1.479, p = 0.139$) shows that the typical (median) engagement ratio did not shift significantly between years. Looking at the distribution curve plots, the curves for both years are centred at approximately the same location, which is why the Mann-Whitney test finds no significant median difference. However, the shape of the curves is dramatically different. The 2025 curve is wide and flat, with a very long right tail extending toward 14.0; the 2026 curve is narrower and taller, concentrated in the range 1 to 3. This visual difference is formally confirmed by the Folded F-test: $F = 3.834, df = [30, 20], p = 0.0027$. The 2025 variance is 3.83 times larger than the 2026 variance, and this ratio is highly statistically significant. It would arise by chance in fewer than 3 of every 1,000 comparisons. The Levene test ($W = 0.075, p = 0.786$) is not

significant, but Levene’s test is robust to outliers by design and tends to understate variance differences when a single extreme value (ratio = 14.0) dominates one distribution. The Folded F-test is more sensitive here and is the appropriate choice. The bootstrap confidence intervals on CV ([24.5%, 218.7%] in 2025 versus [28.5%, 80.4%] in 2026) do not overlap: the 2026 upper bound of 80.4% falls below the 2025 point estimate of 115.9%, providing additional confirmation of the variance reduction. This is a formally and statistically confirmed data quality improvement: the PHS engagement ratio is being measured with significantly greater consistency in 2026 than in 2025.

Year	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
2025		31	2.0588	2.3863	0.4286	1.0000	14.0000
2026		21	2.0996	1.2187	0.2659	0.8286	5.0000
Diff (1-2)	Pooled		-0.0407	2.0027	0.5660		
Diff (1-2)	Satterthwaite		-0.0407		0.5044		

Year	Method	Mean	95% CL Mean	Std Dev	95% CL Std Dev
2025		2.0588	1.1835 2.9341	2.3863	1.9069 3.1897
2026		2.0996	1.5448 2.6543	1.2187	0.9324 1.7599
Diff (1-2)	Pooled	-0.0407	-1.1776 1.0961	2.0027	1.6757 2.4895
Diff (1-2)	Satterthwaite	-0.0407	-1.0554 0.9739		

Method	Variances	DF	t Value	Pr > t
Pooled	Equal	50	-0.07	0.9429
Satterthwaite	Unequal	47.08	-0.08	0.9360

Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	30	20	3.83	0.0027

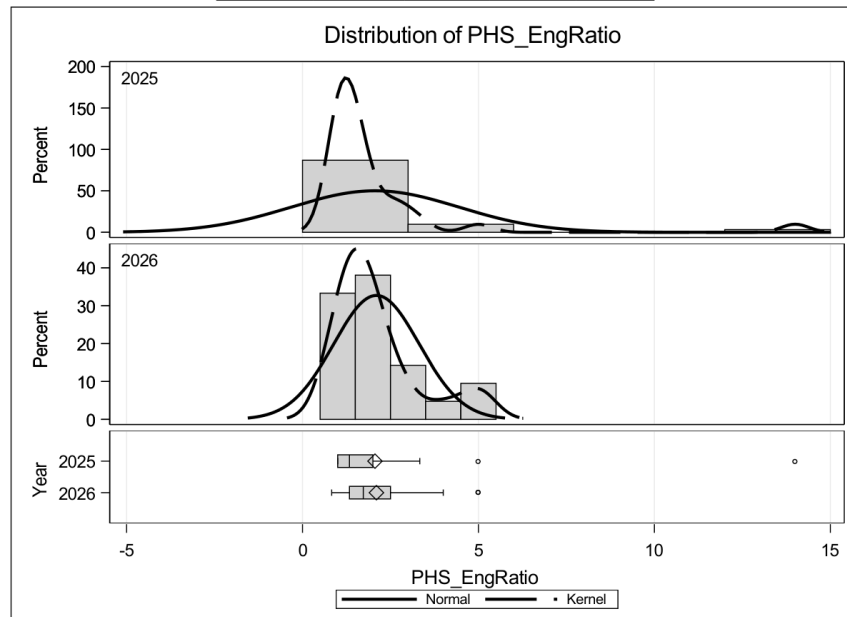


Figure 10: PHS Engagement Ratio distribution curves by year (SAS PROC TTEST, page 97). The 2025 curve (upper panel) is wide and flat. The 2026 curve (lower panel) is narrower and taller. $F = 3.834$, $p = 0.003$ confirms this narrowing is statistically significant.

5.6.3 CH Screening Ratio and OOR Rate

Table 22: CH Screening Ratio and OOR Rate Summary Statistics (SAS PROC MEANS, pages 99, 102).

Statistic	ScrRatio 2025 (<i>n</i> = 29)	ScrRatio 2026 (<i>n</i> = 13)	OOR Rate 2025 (<i>n</i> = 29)	OOR Rate 2026 (<i>n</i> = 13)
Mean	3.334634	4.435897	0.8668	1.2840
Median	2.000000	4.000000	0.6538	1.3333
SD	2.803281	1.937661	0.7309	0.7305
CV	84.0656%	43.6814%	84.3194%	56.8959%
Min	1.000	1.667	0.1667	0.0000
Max	8.000	8.000	2.5526	2.2857

Max = 8.000 both years confirms the eight-test panel. OOR counts test results, not individuals.

Table 23: Mann-Whitney Tests: CH Screening Ratio and OOR Rate (SAS PROC NPAR1WAY, pages 100, 103).

Statistic	Scr Ratio	OOR Rate	What it tells us
Wilcoxon <i>W</i>	350.0	346.0	
<i>Z</i> score	1.9325	1.7961	
<i>p</i> (two-sided)	0.0533	0.0725	Both borderline; just above 0.05
Effect size <i>r</i>	0.374	0.353	Medium for both
95% CI on <i>r</i>	[0.068, 0.616]	[0.045, 0.599]	Entirely above zero for both

Read Me, Community Health tab: “ Number of out of range/positive health screenings. ”

The out-of-range column is fully resolved by this written definition: it counts individual screening results, not individual people, and is analytically usable once that distinction is understood.

Interpretation: Both *p*-values are just above the 0.05 significance threshold (0.053 and 0.073 respectively), making these borderline results. They do not quite reach conventional significance, but the combined evidence from the effect sizes and confidence intervals tells a coherent story. The effect size $r = 0.374$ for the screening ratio and $r = 0.353$ for the OOR rate both fall in the medium range. More importantly, both 95% confidence intervals lie entirely above zero ([0.068, 0.616] and [0.045, 0.599]), meaning we can be 95% confident the true direction of change is upward for both variables. The Wilcoxon score distribution plots show the 2026 boxes sitting above the 2025 boxes in both cases, with some overlap consistent with medium effects. The screening ratio increase from a median of 2.0 to 4.0 means that the typical person screened in 2026 received twice as many individual tests as the typical person in 2025. The minimum of 1.6667 in 2026 (versus 1.000 in 2025) shows that even the events with the fewest tests per person were more comprehensive than in the prior year. The maximum of 8.000 in both years confirms the eight-test panel ceiling is being reached at some

events and that the variable is recording correctly. The OOR rate increase from mean 0.867 to 1.284 means the average person screened in 2026 had approximately 1.3 abnormal results compared to 0.87 in 2025. An OOR rate above 1.0 means more out-of-range test results were recorded than the number of people screened at that event – this is possible and expected because OOR counts individual test results, not individual people. The OOR minimum of 0.000 in 2026 (versus 0.167 in 2025) reflects events where no abnormal results were found, which is a legitimate outcome. The increase in OOR rate warrants monitoring in future data cycles but most likely reflects a combination of more comprehensive testing per person and, potentially, a shift toward events serving higher-need populations.

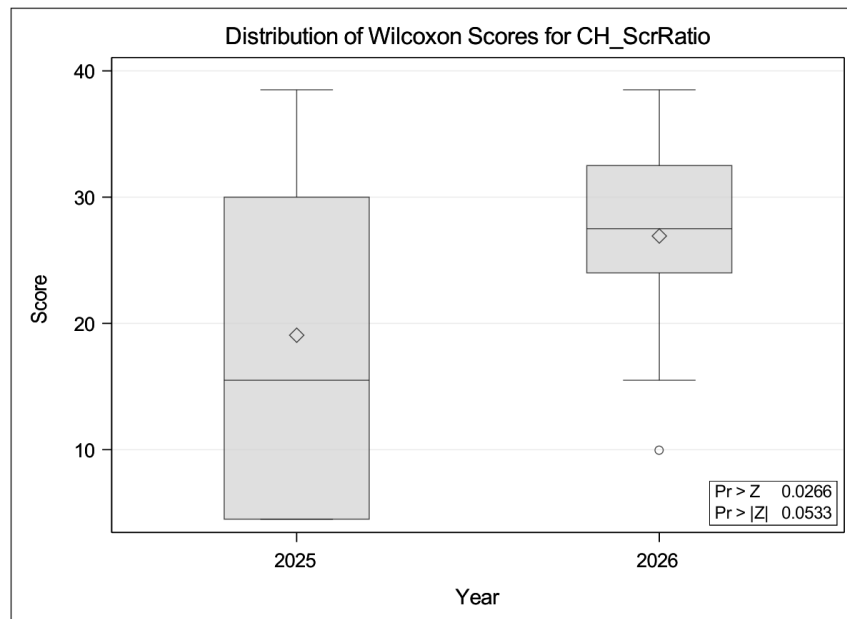


Figure 11: Wilcoxon score distribution: CH Screening Ratio (SAS PROC NPAR1WAY, page 101). The 2026 box sits above the 2025 box (mean rank 26.9 vs 19.1). $p = 0.053$: borderline.

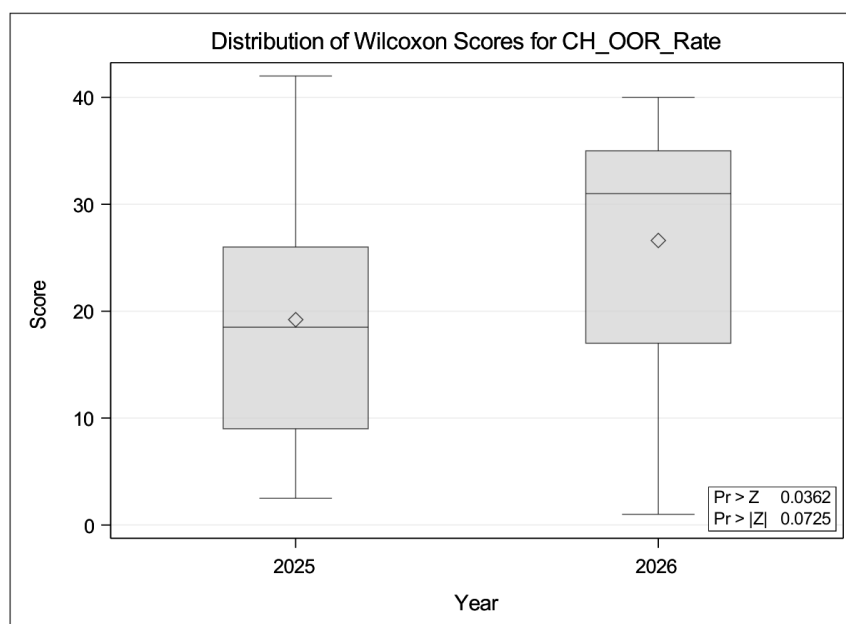


Figure 12: Wilcoxon score distribution: CH OOR Rate (SAS PROC NPAR1WAY, page 104). The 2026 box sits above the 2025 box. $p = 0.073$: borderline. The 95% CI on r does not include zero.

5.6.4 Staff Hours versus Event Hours (SAS PROC MEANS, pages 106–108)

The staff hours to event hours ratio measures how many hours staff collectively logged per hour the event ran. A ratio of 4.0 means that for every hour of event time, staff together logged four hours – consistent with four staff each logging the full event duration. This ratio is a useful staffing intensity measure within each division but cannot be compared across divisions because of the documentation conflict: EH includes travel time in its staff hours while PHS and CH record onsite time only.

Table 24: Staff Hours vs Event Hours by Division and Year (SAS PROC MEANS, pages 106–108). Hours Ratio = PHS/CH/EH Staff Hours ÷ Event Hours.

Division	Year	<i>N</i>	Mean Staff Hrs	Mean Event Hrs	Mean Ratio	Median Ratio	SD Ratio	Max Ratio
PHS	2025	34	14.662	3.156	4.664	4.500	1.436	8.000
PHS	2026	22	12.136	3.023	4.036	4.000	1.059	6.000
CH	2025	30	6.583	4.333	1.548	1.267	0.847	3.000
CH	2026	12	7.769	3.500	2.083	2.000	1.084	4.000
EH	2025	10	11.700	3.045	3.915	3.600	1.447	6.667
EH	2026	1	18.000	6.000	3.000	3.000	n/a	3.000

Critical note: EH Read Me includes travel time in staff hours; PHS and CH Read Me specify onsite time only. Cross-division ratios are not directly comparable. EH 2026: $n = 1$; SD is undefined for a single observation.

Interpretation: For PHS, a mean ratio of 4.66 in 2025 and 4.04 in 2026 means that for

every hour an event ran, PHS staff collectively recorded approximately four hours of work. This is consistent with three to five clinical staff attending each event and each logging their individual time. The ratio is stable across years, which tells us the staffing model has not changed materially. The SD of the ratio declined slightly from 1.44 to 1.06, indicating more consistent staffing levels per event in 2026. For CH, the mean ratio is lower (1.55 in 2025, 2.08 in 2026) because CH typically sends fewer staff per event. The 2026 increase in the CH ratio is consistent with CH attending larger, more complex health screening events where more staff are deployed. For EH, the 2025 mean ratio of 3.92 includes travel and vehicle preparation time per the main Read Me instruction, making it structurally higher than the PHS and CH figures even before accounting for staffing levels. The single 2026 EH event shows 18 staff hours for a 6-hour event (ratio = 3.0), reflecting two staff members including travel. Until the staff hours documentation conflict is resolved, no cross-division comparison of these ratios should be presented to leadership.

5.6.5 Spearman Correlations: Does Staffing Predict Service Volume?

A Spearman correlation (r) measures whether two variables move together consistently. $r = +1.0$ is a perfect positive relationship; $r = 0$ is none.

Table 25: Spearman Correlations (SAS PROC CORR, pages 110–112). 95% CI via Fisher z -transformation.

Relationship	r	95% CI	p	Sig.	n
CH staff → CH screenings	0.68251	[0.471, 0.820]	< 0.0001	***	40
CH staff → CH engagements	0.44482	[0.163, 0.660]	0.0032	**	42
Event hours → CH engagements	0.38881	[0.092, 0.622]	0.0120	*	41
PHS staff → STI tests	0.56511	[0.286, 0.756]	0.0004	***	35
PHS staff → people served	0.32520	[0.071, 0.540]	0.0136	*	57
PHS vs. CH engage (co-attend)	0.59895	[0.183, 0.833]	0.0086	**	18
PHS staff → engagements	0.25697	[−0.004, 0.485]	0.0537	ns	57
*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; ns = not significant.					

Interpretation: The Spearman correlation matrix reveals consistent, statistically confirmed relationships between staffing levels and service output across both divisions. For CH, the correlation between staff count and screenings delivered ($r = 0.683$, $p < 0.001$, $n = 40$) is the strongest relationship in the dataset. The 95% CI [0.471, 0.820] lies entirely above 0.4, confirming a large effect even in the most conservative scenario. This means that events where CH deployed more staff consistently produced substantially more health screenings – the relationship is not coincidental and is not driven by a few outlier events. The correlation between CH staff and engagements ($r = 0.445$, $p = 0.003$) and between event hours and CH engagements ($r = 0.389$, $p = 0.012$) both confirm that both staffing intensity and event duration independently predict CH service volume. For PHS, the relationship between staff count and STI tests delivered ($r = 0.565$, $p < 0.001$) is also large and highly significant. The relationship between PHS staff and people served ($r = 0.325$, $p = 0.014$) is smaller

but still significant, suggesting staff count has a stronger influence on STI testing volume than on general service reach. The correlation between PHS and CH engagements at co-attended events ($r = 0.599$, $p = 0.009$, $n = 18$) is particularly informative: when the same event draws high PHS engagement, it also draws high CH engagement. The two divisions appear to be responding to the same underlying event characteristics – likely event size and population density – rather than operating independently. The one non-significant relationship, PHS staff and PHS engagements ($r = 0.257$, $p = 0.054$), sits just above the threshold, suggesting a weaker staffing-to-engagement link for PHS that may reflect the more structured, appointment-based nature of PHS clinical interactions. Across the board, adequate staffing at every event is not merely a resource question; it is the single strongest operational predictor of service volume in this dataset.

5.7 5.7 Coherence – All Divisions: MEDIUM

Coherence is the ability to maintain consistent definitions and classification systems across divisions and over time. The MHU tracking sheet is an integrated data product in the FCSM sense: three divisions with different workflows, systems, and populations contribute into one shared workbook. The coherence threats documented here are therefore integration-level threats, not merely division-level inconsistencies, and several of them require a cross-division decision to resolve rather than a single division acting alone.

5.7.1 Threat 1: The Staff Hours Documentation Conflict

This is the single most consequential coherence threat identified in this assessment. The main Read Me tab and the division-specific Read Me tabs give contradictory written instructions for the same variable.

Read Me, Main tab, EH section: “ Total Number of Hours from the time staff left the office to coming back. This includes travel time, time for doing pre/post vehicle checks, time onsite providing services. ”

Read Me, PHS and CH tabs: “ Hours staff were onsite during the event. Do NOT include travel time or pre/post vehicle checks. ”

All three divisions have been following their respective written instructions correctly. This is not a staff error. It is a documentation conflict embedded in the workbook itself, and it directly explains why the staff hours ratios in Section 5.6.4 are not comparable across divisions.

Joint Display: *Divergence, formally resolved*

Table 26: Joint Display 1: Staff Hours Documentation Conflict

Strand	Evidence	What it establishes
Qualitative	Read Me, Main tab (EH): total hours including travel. Read Me, PHS/CH tabs: onsite hours only. Two contradictory written instructions in the same workbook.	The contradiction is a documentation defect, not a staff behaviour problem; both instructions were followed exactly as written.
Quantitative	SAS PROC MEANS (pp. 106–108): EH mean hours ratio = 3.92 (2025) vs PHS = 4.66 and CH = 1.55.	The numeric gap between divisions cannot be interpreted as a true staffing-intensity difference, because the underlying variable is not measuring the same construct across divisions.

Integration outcome: divergence, formally resolved. The qualitative strand identifies *why* the quantitative ratios are not comparable; the quantitative strand confirms the size of the resulting gap is large enough to matter (PHS ratio is 3.0× the CH ratio). Neither strand alone would have caught this: the Read Me text alone does not show the magnitude of consequence, and the ratios alone would have been read as a real staffing difference without the Read Me.

5.7.2 Threat 2: Engagement Definition Drift

As established in Section 3, the written standard defines engagement as the number of interactions, not the number of people. PHS and EH both count individuals instead, while CH practice aligns most closely with the written definition.

5.7.3 Threat 3: Schema Changes Between 2025 and 2026 (Undocumented)

Comparing both workbook column lists directly reveals that nine new PHS columns were added in 2026, including health screenings, Narcan, A1C, and CIIS record searches, none of which were documented through any formal change log. The CH staff columns were also relabelled from *Number of CH Staff (non-CHW)* and *Number of CH Staff Hours (non-CHW)* to *Number of CH Staff* and *Number of CH Staff Hours*, silently removing the CHW exclusion language. Anyone comparing 2025 and 2026 CH staff counts without knowing about this relabelling would be comparing two different things without realising it.

This kind of undocumented schema drift is consistent with how the supervisor describes the tracking sheet's origins: it was never built against a fixed specification, but assembled incrementally as the programme's needs changed.

Public Health Services, project inception meeting, lightly edited for clarity: “This is not a plug and play. This is a design build. Nobody told us how to do this. We had to build it from scratch. There was no out-of-the-box data system we could grab, or even if there was, we didn’t get the money to buy it. So we tried to do this kind of mosaic of spreadsheets.”

This context does not excuse the lack of a change log, but it explains why one never existed in the first place: the tracking sheet was treated as a working operational tool that could be edited as needs arose, not as a fixed instrument whose structure had to be version-controlled. Recommendation R13 addresses this going forward.

5.7.4 Threat 4: Regional Coverage Bias

As the Kruskal-Wallis test below confirms quantitatively, PHS engagement is heavily concentrated in Greeley/Evans. Any spatial or regional analysis built on this dataset carries a geographic bias that needs to be disclosed alongside the analysis, not discovered by the reader independently.

5.7.5 Threat 5: Inconsistent EH Attendance Encoding

EH records attendance using a three-level field (Outreach Only; Testing and Outreach; Did not attend) while PHS and CH use a binary yes/no field. Any cross-division attendance comparison requires a transformation step that is not currently documented anywhere in the tracking sheet or its Read Me tabs.

5.7.6 Threat 6: Unique Event Identifier Absent

No unique event identifier exists anywhere in the tracking sheet. When the same physical location is visited more than once, Tableau’s distinct count functions produce errors because there is no reliable key to disambiguate repeat visits.

***Data Team:** “ The same address used multiple times per quarter creates distinct count errors. I manually went in and edited the column in the tracking sheet so the string would not count it twice. That is just going to get more common as we go to locations on a recurring basis. ”*

This manual string-editing workaround is technically functional but creates its own accuracy risk: an edited address no longer matches the original source record, and there is no log of which addresses have been edited or why. Recommendation R12 addresses this directly.

5.7.7 Threat 7: CH Engagement Apparent Decline

The raw CH engagement total fell from 692 in 2025 to 439 in 2026. As demonstrated quantitatively below and revisited in Cross-Cutting Finding 6.5, this is an artefact of the unequal observation windows rather than a real decline, but it is exactly the kind of figure that could be misread by anyone unfamiliar with the underlying data structure if it were extracted from the tracking sheet without this context.

5.7.8 Threat 8: Decentralised Entry With No Cross-Division Verification

The tracking sheet is not entered or maintained by a single person or team. Each division is independently responsible for completing its own section of the shared sheet, and division leads who fill in their own portion explicitly do not know how other divisions complete theirs.

***Public Health Services, lightly edited for clarity:** “ I don’t know how many services community health is, I don’t know how many screenings lead does, until we go in the meetings. ”*

The data team’s role is limited to producing the original formatting (dropdown menus, whole-number validation) and pulling the data divisions have already entered; they do not enter data themselves and do not verify it against any source record before using it.

***Data Team, lightly edited for clarity:** “ We don’t enter data into the tracking sheet. We just ask them to do that and we trust that that happens, and then we just pull it. ”*

The only verification mechanism across all three divisions is a single quarterly reminder email asking each division to update the sheet by a fixed date, with no follow-up to confirm the request was acted on. This means the tracking sheet’s overall coherence depends entirely on each division correctly understanding and applying the shared Read Me definitions on their own, without any cross-division review step at any point before the data reaches SAS for analysis.

5.7.9 Regional Analysis: Kruskal-Wallis Test

The Kruskal-Wallis test asks whether per-event PHS engagement volumes differ across the five geographic regions. H is the test statistic; epsilon-squared (ε^2) is the effect size: 0.01 = small, 0.06 = medium, 0.14 = large.

Table 27: Kruskal-Wallis: PHS Engagements by Region (SAS PROC NPAR1WAY, page 113; descriptive statistics from SAS PROC MEANS, page 114). $\chi^2 = 17.2433$, $df = 4$, $p = 0.0017$. $\varepsilon^2 = 0.263$ – Large.

Region	N Events	Wilcoxon Sum	Mean Score	Median	Mean
Greeley/Evans	41	1419.50	34.621951	12.0	17.49
North	4	74.00	18.500000	6.5	7.25
Johnstown/Milliken	1	18.50	18.500000	7.0	7.00
Southeast	5	65.50	13.100000	5.0	5.40
Windsor/Severance	6	75.50	12.583333	5.5	5.17

Interpretation: The Kruskal-Wallis test yields $H = 17.243$, $df = 4$, $p = 0.0017$, which is highly significant. Looking at the Wilcoxon score distribution plot, the Greeley/ Evans box is visibly separated from all other regions: its lower quartile (25th percentile) sits at approximately the same height as the upper quartiles (75th percentile) of the Southeast and Windsor/Severance regions, indicating that even the events with below-average PHS engagement in Greeley/Evans performed as well as or better than the best-performing events in those regions. The mean score for Greeley/ Evans (34.6) is more than 2.5 times higher than the next highest region (North, 18.5), and the Wilcoxon sum for Greeley/Evans (1,419.50) accounts for the vast majority of all rank points, which is consistent with 41 of the 57 events with PHS presence being held in Greeley/Evans. Epsilon-squared = 0.263 is the non-parametric effect size equivalent of eta-squared: it means that region membership alone explains 26.3% of all variation in PHS engagement scores across events. By Cohen’s benchmarks for epsilon-squared, a value above 0.14 is large; 0.263 represents a very large effect. The Dunn post-hoc test with Bonferroni correction identifies the specific source of

the overall difference: Greeley/Evans versus Windsor/Severance ($z = 3.021$, $p_{\text{Bonf}} = 0.015$) and Greeley/Evans versus Southeast ($z = 2.740$, $p_{\text{Bonf}} = 0.037$) are the only statistically significant pairwise contrasts. All comparisons not involving Greeley/Evans are non-significant, confirming that the four smaller regions are broadly similar to each other but together constitute a group that performs at substantially lower PHS engagement than Greeley/Evans. The practical implication is that PHS engagement is geography-dependent in a way that CH engagement is not. Understanding why Greeley/Evans events produce substantially more engagement interactions per event – whether due to population density, event type, venue characteristics, or staffing patterns – would be a productive focus for programme optimisation.

Wilcoxon Scores (Rank Sums) for Variable PHS_Engage Classified by Variable Region					
Region	N	Sum of Scores	Expected Under H0	Std Dev Under H0	Mean Score
North	4	74.00	116.0	31.935114	18.500000
Greeley/Evans	41	1419.50	1189.0	56.176213	34.621951
Johnstown/Milliken	1	18.50	29.0	16.413249	18.500000
Southeast	5	65.50	145.0	35.366104	13.100000
Windsor/Severance	6	75.50	174.0	38.367301	12.583333

Average scores were used for ties.

Kruskal-Wallis Test	
Chi-Square	17.2433
DF	4
Pr > Chi-Square	0.0017

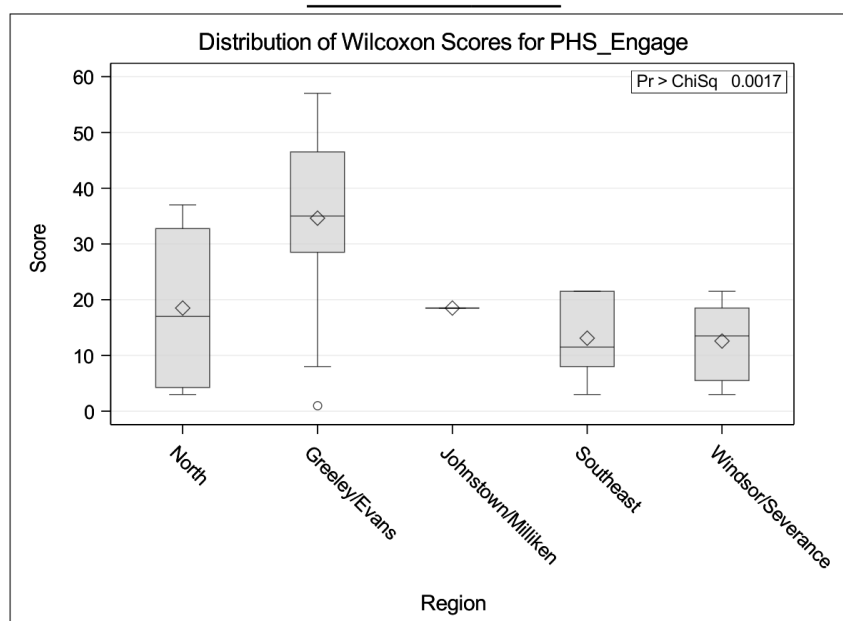


Figure 13: Kruskal-Wallis: PHS Engagements by region (SAS PROC NPAR1WAY, page 113). The Greeley/Evans box sits clearly above all others. The bottom of the Greeley/Evans box is higher than the tops of the Southeast and Windsor/Severance boxes. $Pr > \chi^2 = 0.0017$ in the upper right corner confirms high significance.

Table 28: Dunn Post-hoc Test with Bonferroni Correction, 10 comparisons (SAS PROC NPAR1WAY, page 115).

Comparison	z	p (raw)	p (Bonferroni)	Significant
Greeley/Evans vs Windsor/Severance	3.021	0.0025	0.0151	Yes
Greeley/Evans vs Southeast	2.740	0.0061	0.0369	Yes
Greeley/Evans vs North	1.856	0.0635	0.3807	No
North vs Southeast	0.486	0.6272	1.0000	No
North vs Windsor/Severance	0.540	0.5895	1.0000	No
Southeast vs Windsor/Severance	0.037	0.9703	1.0000	No

Wilcoxon Scores (Rank Sums) for Variable CH_Engage Classified by Variable Region					
Region	N	Sum of Scores	Expected Under H0	Std Dev Under H0	Mean Score
Greeley/Evans	31	645.50	651.0	32.904889	20.822581
Southeast	6	131.50	126.0	27.082530	21.916667
North	1	2.50	21.0	11.819785	2.500000
Windsor/Severance	2	52.00	42.0	16.505431	26.000000
Johnstown/Milliken	1	29.50	21.0	11.819785	29.500000

Average scores were used for ties.

Kruskal-Wallis Test	
Chi-Square	3.2857
DF	4
Pr > Chi-Square	0.5112

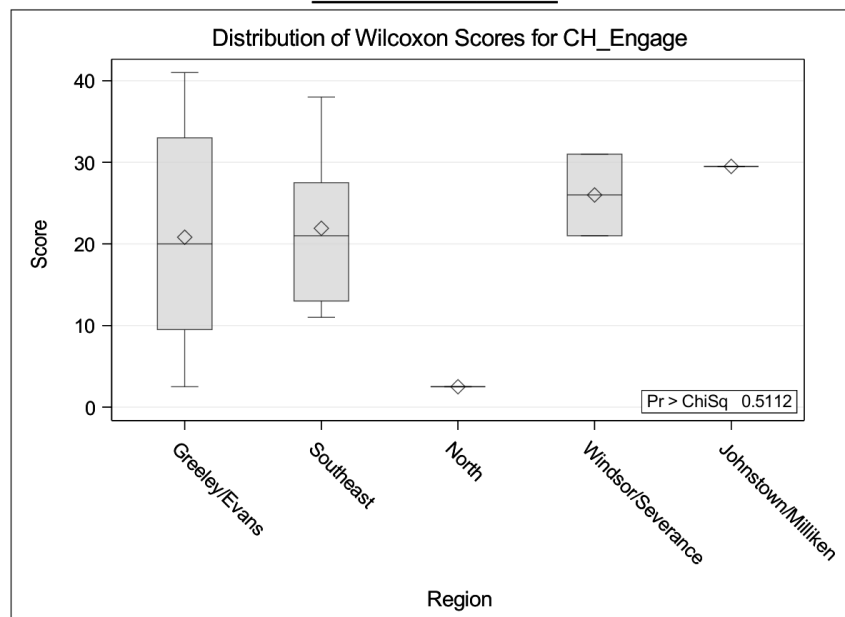


Figure 14: Kruskal-Wallis: CH Engagements by region (SAS PROC NPAR1WAY, page 116). The boxes across all regions overlap substantially. $H = 3.286$, $p = 0.511$: no significant regional difference. CH delivers consistently regardless of geography, unlike PHS.

5.7.10 Address Completeness

Table 29: Address Completeness by Year (SAS PROC FREQ, page 117). $N = 85$; no missing.

Year	Complete (Addr = 1)	Total	% Complete
2025	37	54	68.52%
2026	7	31	22.58%

Table 30: Statistical Tests for Address Completeness Decline (SAS PROC FREQ, page 117). All four tests agree.

Test	Statistic	<i>p</i> -value	Notes
Pearson χ^2	16.6448	< 0.0001	
Continuity-adjusted (preferred)	14.8559	0.0001	Conservative; recommended
Likelihood Ratio χ^2	17.3381	< 0.0001	
Fisher Exact (two-sided)	–	< 0.0001	
Phi coefficient	–0.4425	–	Medium-large negative effect

Table 31: Logistic Regression: Address Missingness MAR Test (SAS PROC LOGISTIC, page 119). Reference: Greeley/Evans, Year 2025.

Diagnostic / Predictor	Value	95% CI	<i>p</i>	Sig.
McFadden pseudo R^2	0.211			
Hosmer-Lemeshow <i>p</i>	0.948			Excellent fit
C-statistic (AUC)	0.777			
Overall accuracy	75.0%			
Sensitivity / Specificity	81.8% / 67.5%			
Year 2026 vs 2025 (OR)	0.091	[0.029, 0.289]	< 0.001	***
North vs Greeley/Evans	0.105	[0.011, 1.040]	0.054	No
Windsor vs Greeley/Evans	3.894	[0.637, 23.812]	0.141	No
Southeast vs Greeley/Evans	1.744	[0.339, 8.983]	0.506	No

Interpretation: All four chi-square-family tests agree, and the Fisher Exact test gives the same conclusion: the drop from 68.52% complete addresses in 2025 to 22.58% in 2026 is real and highly significant. The continuity-adjusted chi-square ($\chi^2 = 14.856$, $p = 0.0001$) is the preferred statistic for a two-by-two table of this structure as it applies a correction that makes the result more conservative; even with this correction, the *p*-value is far below 0.001. Phi = –0.4425 indicates a medium-to-large negative association: a value of –0.44 on a –1 to +1 scale represents a substantial relationship between year and address completeness. The logistic regression provides the key inferential result. The Hosmer-Lemeshow $p = 0.948$ confirms excellent model fit – a value above 0.05 means the model’s predicted probabilities match the observed outcomes very well. The C-statistic (AUC) of 0.777 means the model correctly discriminates between complete and incomplete addresses 77.7% of the time, which is fair to good discrimination. Overall accuracy of 75.0%, sensitivity of 81.8%, and specificity of 67.5% confirm the model is performing well. The critical finding is the odds ratio for year: OR = 0.091, 95% CI [0.029, 0.289], $p < 0.001$. An odds ratio of 0.091 means the odds of having a complete address in 2026 are 90.9% lower than in 2025 after controlling for region. No region is statistically significant, and all regional confidence intervals include 1.0. Year is the single dominant predictor of address missingness. This pattern is defined as MAR

(Missing At Random): the missing data is predictable from an observed variable (year) rather than from the missing values themselves. MAR is important because it means the cause of missingness is identifiable and in principle fixable – the programme can investigate what changed between 2025 and 2026 in how addresses were collected or recorded, and correct the process for future events.

This is the single largest, most statistically confirmed decline in the entire dataset, and the qualitative record does offer a partial explanation, though not yet a complete one. In the project inception meeting, the supervisor’s working explanation was that the gap reflects a staff education issue rather than a system failure: staff have not been given a clear reason why complete addresses matter for the programme, so the field gets treated as low priority during busy events. Address quality was also written into every division’s interview guide as a priority question, specifically asking who is responsible for entering the address and why completeness declined in 2026 rather than improved, but those interviews had not yet produced a confirmed answer at the time of this report.

The MAR result in the logistic regression above is consistent with a staff education or prioritisation explanation: if address entry is being deprioritised under time pressure, that pressure would plausibly differ by year (matching the strong year effect) without being explained by region, which is exactly the pattern the regression shows. That said, a working explanation is not a confirmed one, and this remains the clearest example in the report where the quantitative evidence is stronger than the qualitative evidence behind it. Confirming or ruling out the staff education explanation directly with frontline staff, rather than relying on a single supervisory account, is the specific purpose of Recommendation R10.

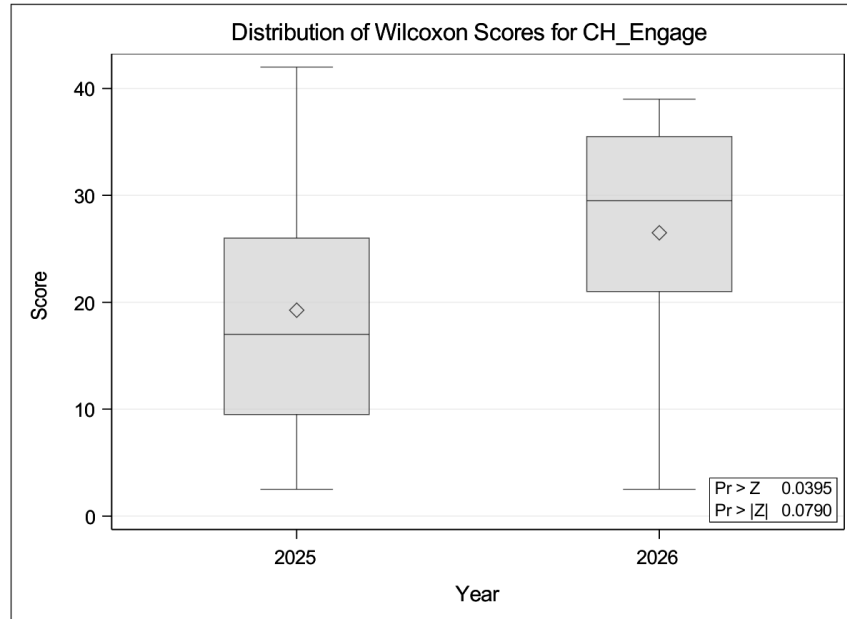


Figure 15: CH Engagements per event: coherence check (SAS PROC NPAR1WAY, page 121). $W = 344.5$, $Z = 1.7567$, $p = 0.079$. The 2026 box sits above the 2025 box (mean rank 26.5 vs 19.3), confirming the per-event rate is rising despite the raw total decline. Annualised: 1,038 (2025) vs 1,054 (2026).

5.8 Scientific Integrity – All Divisions: MEDIUM-HIGH

Scientific Integrity is the use of established standards and professional methods to collect and report data.

Table 32: Coefficient of Variation Analysis (SAS PROC MEANS, pages 124–126). Bootstrap 95% CI over 5,000 resamples.

Variable	Yr	N	Mean	SD	CV	Bootstrap 95% CI	Variance test
STI Ratio	2025	15	5.2258	1.2109	23.17%	[11.1%, 28.7%]	Levene $p = 0.293$ ns
STI Ratio	2026	17	5.8302	0.8213	14.09%	[5.2%, 20.2%]	F-test $p = 0.138$ ns
PHS Eng. Ratio	2025	31	2.0588	2.3863	115.90%	[24.5%, 218.7%]	Levene $p = 0.786$ ns
PHS Eng. Ratio	2026	21	2.0996	1.2187	58.05%	[28.5%, 80.4%]	F-test $p = 0.003$ *
CH Scr. Ratio	2025	29	3.3346	2.8033	84.07%	[54.1%, 118.1%]	–
CH Scr. Ratio	2026	13	4.4359	1.9377	43.68%	[25.9%, 57.6%]	–

* F-test: $F = 3.834$, $df = [30, 20]$, confirmed from SAS PROC TTEST, page 97.

PHS bootstrap CIs do not overlap: 2026 upper bound (80.4%) is below the 2025 estimate (115.9%).

Interpretation: The CV table shows a consistent directional trend across all three ratio

variables: every CV declined from 2025 to 2026, indicating more consistent application of each measurement across events. However, the strength of the statistical evidence differs. For the STI ratio, CV declined from 23.17% to 14.09%, a reduction of nine percentage points. Both variance tests are not significant (Levene $p = 0.293$, F-test $p = 0.138$), and the bootstrap CIs overlap ([11.1%, 28.7%] versus [5.2%, 20.2%]), so this improvement is directional and encouraging but not yet formally statistically confirmed at the current sample size of 15 and 17 events. For the PHS engagement ratio, the evidence is much stronger. CV declined from 115.9% to 58.0%, a reduction of 57.9 percentage points. The F-test is statistically significant ($F = 3.834$, $p = 0.003$), and the bootstrap CIs do not overlap (the upper bound of the 2026 interval, 80.4%, falls below the 2025 point estimate of 115.9%). Both forms of evidence confirm the variance reduction is real. For the CH screening ratio, CV declined from 84.07% to 43.68%. No formal variance test was run for this variable in the SAS output, but the bootstrap CIs do not overlap: the 2026 upper bound of 57.6% falls below the 2025 lower bound of 54.1%, providing strong informal evidence that the improvement is genuine. The high CV in both years for CH is expected and appropriate: CH offers a panel of eight different tests and each event deploys a different combination depending on the anticipated population, so a ratio measured at the event level will legitimately vary across events in a way that does not indicate inconsistency. The most important single finding in this section is the Read Me documentation itself: written, agreed definitions for all key data terms produced through group discussion before operations began is rare in first-year public health outreach programmes, and it forms the foundation that makes data quality improvement detectable and sustainable.

The Read Me documents are the single most important positive finding in this assessment. Comprehensive written definitions for all key data terms, established through group discussion before operations began and confirmed independently by all four interview groups, is rare in first-year public health outreach programmes.

Community Health: “ We all sat down and determined how each of these are counted so there is consistency. It is written in the Read Me tab. ”

Data Team: “ We spent a lot of time discussing this as a group. ”

The rating is Medium-High rather than High for three specific reasons. First, the engagement definition has drifted from the written standard in practice for PHS and EH, as documented in Coherence Threat 2. Second, no EH-specific Read Me tab exists in either workbook, meaning EH entered the programme without access to the same division-level written guidance available to PHS and CH. Third, the CH health screenings Read Me lists only four tests while current clinical practice includes eight, an undocumented expansion of the screening panel that is not reflected anywhere in the written guidance.

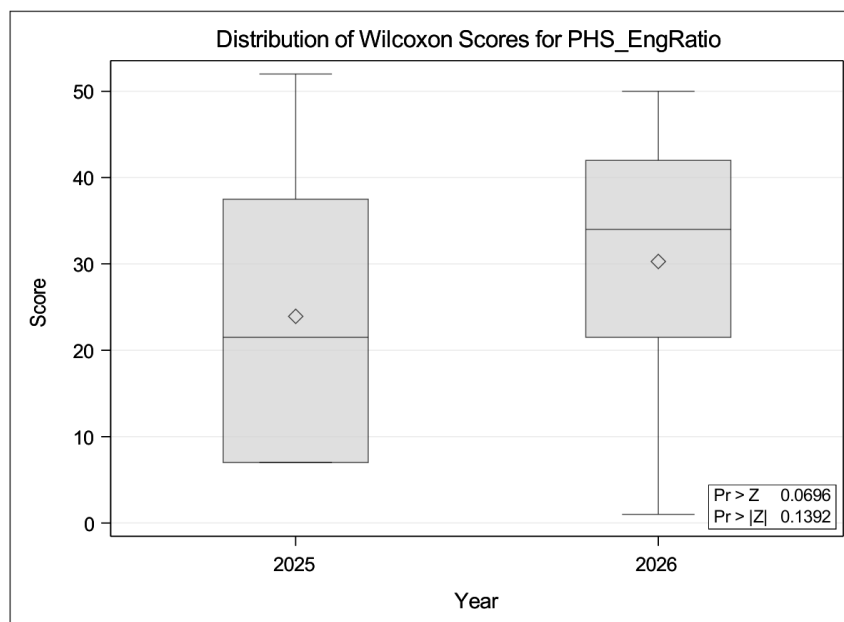


Figure 16: Wilcoxon score distribution: PHS Engagement Ratio (SAS PROC NPAR1WAY, page 128). The two boxes sit at similar heights, confirming similar medians ($p = 0.139$). The 2026 box is visibly narrower, capturing the reduced variance. The F-test ($p = 0.003$) confirms this narrowing is statistically significant.

5.9 5.9 Credibility – All Divisions: HIGH

Credibility reflects confidence in data based on the track record and methods of those who produce it.

Public Health Services: “ I have to justify it. And the best way to do that is with the data. Numbers do not lie. They could be manipulated, but they do not lie. ”

Credibility is rated High across all three divisions. Every pattern in the tracking sheet that initially appeared anomalous during this assessment was investigated and confirmed as intentional through interviews and Read Me cross-reference, rather than being left as an unexplained irregularity.

Two potential anomalies that emerged from quantitative analysis were investigated qualitatively and resolved. First, the DPP data gap: Community Health’s Diabetes Prevention Programme data does not appear in the tracking sheet, and the logical question was whether this represented a gap.

Community Health: “ The screenings are a recruitment tool for our DPP classes. CDC gets the DPP data separately – it has nothing to do with the mobile unit tracking sheet. ”

This is a legitimate programme design decision, not a data gap. Second, the CH out-of-range

count: the OOR column records results, not people, which initially appeared anomalous, and Community Health explained precisely how counts are handled.

Community Health: “ We write down the results for the client. We keep a tally mark that that person was out of range. We do not keep their individual health information. ”

For EH, the mandatory reporting requirement for blood lead tests means every test result is recorded independently in a state system regardless of what happens to the tracking sheet.

Environmental Health: “ All information on the consent form is reported to EpiTrax since blood lead test results are mandatorily reportable. ”

5.9.1 STI Ratio All 32 Events Combined

Table 33: STI Ratio: All 32 Events Combined (SAS PROC UNIVARIATE, pages 129–132). $N = 32$ includes all STI events from 2025 and 2026.

Statistic	Value (from SAS)	Note
N	32	All STI events combined
Mean	5.5469292	
Median	6.0000000	Exactly at the six-test standard
Mode	6.0000000	Most common value equals the standard
SD	1.0508079	
CV	18.9440%	
SE Mean	0.1857584	
95% CI on mean	[5.1681, 5.9258]	All values above 5.0
Skewness	−0.7375166	Slight left tail from early events
Kurtosis	2.8480618	
IQR	0.7750	50% of events within 0.775 of median
Q_1 / Q_3	5.2250 / 6.0000	Upper quartile equals the standard
Range	[2.6667, 8.4000]	

Interpretation: The combined STI ratio across all 32 events presents the strongest single piece of evidence for the credibility of PHS data recording. The median and mode are both exactly 6.000, meaning the most common single value and the middle value in the entire two-year dataset both equal the written standard precisely. In the histogram, the tallest bar sits at 6.0, which is the visual confirmation: more events produced a ratio of exactly 6.0 than any other value. The IQR of 0.775 means the middle 50% of all 32 events fell within a range of only 0.775 tests per person around the median, which is a narrow band for a variable measured across events of different sizes and compositions. The 95% CI on the mean [5.168, 5.926] lies entirely above 5.0 and includes 6.0, so the average is statistically consistent with the standard. The mean of 5.547 is slightly below 6.0, and the slight left skew (skewness = −0.737) reflects the small number of early events that produced lower ratios before the six-test protocol was fully standardised across all staff – these pull the mean

modestly downward without affecting the central tendency shown by the median and mode. The kurtosis of 2.848 is above the normal distribution value of 0, indicating a somewhat peaked distribution, which is consistent with values concentrating near 6.0 with a small number of outliers in the tails. The normality tests (KS $p < 0.010$; CvM $p < 0.005$; AD $p < 0.005$) all reject normality, confirming that the distribution concentrates at a fixed value rather than spreading like a bell curve – exactly what we expect from a standardised protocol. Taken as a whole, this analysis confirms that the six-test STI standard is a real, consistently applied operational protocol rather than a target that is loosely followed.

Joint Display: *Convergence*

Table 34: Joint Display 4: The Six-Test STI Standard

Strand	Evidence	What it establishes
Qualitative	PHS staff describe in interview a written six-test STI panel applied at every STI event, confirmed independently across all PHS interview participants without prompting.	Staff report the standard is followed consistently, in their own account of practice.
Quantitative	SAS PROC UNIVARIATE (p. 129): median = 6.000, mode = 6.000 across all 32 STI events; IQR = 0.775.	The tracking sheet data independently confirms the same pattern the staff described, with no prompting from the data toward that conclusion.

Integration outcome: convergence. This is the strongest credibility finding in the report precisely because the two strands were collected independently and arrived at the same answer without either one informing the other in advance.

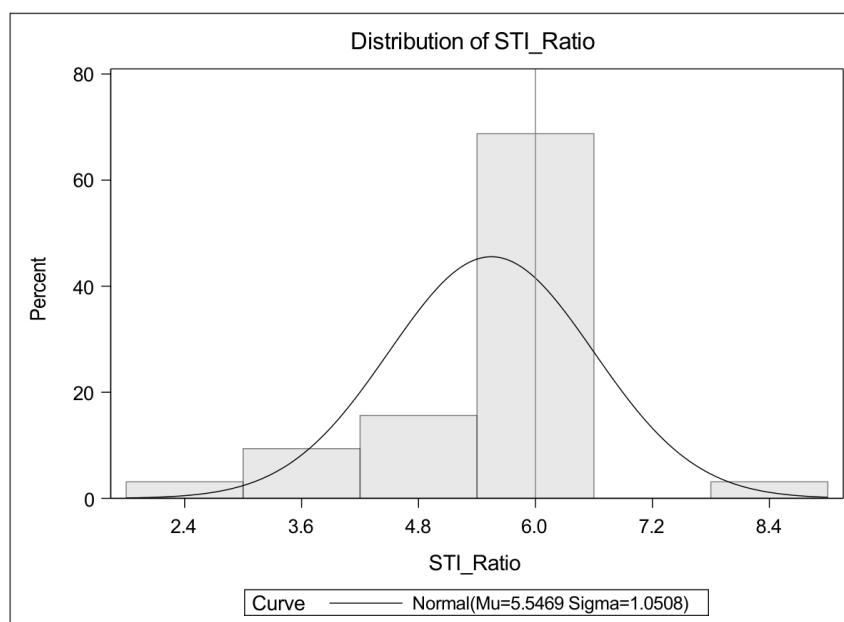


Figure 17: STI Ratio distribution, all 32 events combined (SAS PROC UNIVARIATE, page 131). The tallest bar sits at 6.0. The smooth curve shows what a normal distribution with the same mean and SD would look like: the data concentrates at the fixed standard rather than spreading symmetrically. Mean = 5.547; Median = 6.000; Mode = 6.000.

5.9.2 Customer Experience Survey

Table 35: Customer Satisfaction by Domain ($N = 315$). Cronbach $\alpha = 0.894$ (GOOD). Scale: 5 = Great, 4 = Good, 3 = OK, 2 = Fair, 1 = Poor. Source: customer experience survey analysis. Domain means and percentages are from a separate SAS run not included in the primary 148-page SAS output (MHU_Phase1_Output.pdf). Cronbach α computed via SAS PROC RELIABILITY on the same dataset.

Domain	N	Mean	% Great	% Great or Good
Overall satisfaction	313	4.94	95.2%	99.0%
Wait time	312	4.84	88.8%	97.4%
Staff respect	313	4.94	94.9%	99.4%
Staff helpfulness	311	4.95	95.8%	99.0%
Staff time with patient	313	4.93	93.6%	98.7%
Staff explanation clarity	314	4.95	95.9%	99.4%
Staff confidentiality	312	4.93	94.2%	97.8%
Location accessibility	314	4.90	93.6%	97.5%
Hours convenience	312	4.91	93.9%	97.4%
Cleanliness	307	4.87	90.6%	93.8%

Table 36: Recommendation and Unmet Need (SAS PROC FREQ, pages 138–139).

Measure	N	Percent	Grant significance
Definitely will recommend	267	84.76%	
Probably will recommend	18	5.71%	
Would NOT get care elsewhere	202	64.13%	Primary unmet need measure
Primary barrier: cost too high	72	35.6% of 202	Direct grant justification
Barrier: do not know how	58	28.7% of 202	
Barrier: no transportation	36	17.8% of 202	
Barrier: inconvenient hours	29	14.4% of 202	
202 of 293 valid respondents (68.9%) had no alternative care source.			

Interpretation: Cronbach $\alpha = 0.894$ is a measure of internal consistency: it confirms that all ten satisfaction items are measuring the same underlying construct (overall patient experience with the MHU) rather than ten unrelated things. A value above 0.80 is conventionally considered acceptable, above 0.90 is excellent; 0.894 sits at the top of the acceptable range and approaches excellent. This means the survey instrument is reliable and the composite satisfaction score can be trusted as a coherent measure. Every domain scores above 88% in the Great category and above 93% in the combined Great or Good category. The highest-rated domains are Staff Helpfulness (4.95 mean, 95.8% Great) and Staff Explanation Clarity (4.95 mean, 95.9% Great), which are both close to the ceiling of the five-point scale. These results indicate that clinical staff are delivering care that patients experience as both effective and respectful. The lowest-rated domain, Cleanliness (4.87 mean, 90.6% Great, 93.8% Great or Good), still scores extremely well in absolute terms; the relative ranking simply reflects the constraints of a mobile unit environment rather than any meaningful quality deficit. The single most powerful grant justification number in this entire assessment is 202 (64.13% of 315 respondents): nearly two out of three patients surveyed reported that they would not have gotten care anywhere else if the MHU had not been present at that event. This figure, drawn from 315 responses across 315 distinct patient contacts, establishes that the MHU is not supplementing or duplicating existing care pathways. For the majority of its patients, the MHU is the sole access point. Among the 202 patients reporting no alternative, the top barrier was cost (35.6%), followed by not knowing how to access care (28.7%), no transportation (17.8%), and inconvenient hours (14.4%). These four barriers together define the target population the MHU is designed to serve, and this survey confirms the programme is reaching them.

5.10 5.10 Computer and Physical Security – All Divisions: MEDIUM

The master tracking sheet is stored in a personal OneDrive account and the risk this creates is not hypothetical. It has already happened. The original file was created and owned by a staff member in PHS, and when that person left the county their OneDrive account was

deactivated, taking the file with it. The data team happened to have a locally saved copy from a prior export, which is the only reason the programme’s historical data survived; that saved copy was used to build the 2026 tracking sheet.

Data Team: “ *Currently, someone has to own something in OneDrive, and currently the division director for our community health, technically it’s inside of her OneDrive. She made it and shared it with everyone. I’m pretty sure it’s her, but I can double check.* ”

Ownership has already moved once: the original owner was a PHS staff member who left the county, and current ownership most likely sits with a different individual in a different division. The data team’s own account of this was explicitly tentative rather than confirmed, and that confirmation was still pending at the time of this report. The risk this report is flagging is therefore not specific to any one individual or division; it is structural. The file has already changed ownership once due to staff turnover, the loss already happened once as a direct result of that turnover, and nothing about the current arrangement prevents the same sequence from repeating with whoever holds the file next.

Environmental Health, written response: “ *I do hope the data is being archived by the data team somewhere because we have had this issue in the past where people leave and the documents on OneDrive are gone. I think OneDrive deletes stuff fairly soon after your account is disabled with the county.* ”

This vulnerability was identified by someone in a different division from the one responsible for the file, which is itself a sign of reactive rather than proactive security awareness across the programme. The fix is straightforward: transfer the file to a shared Data Team account not tied to any individual staff member. This takes approximately 30 minutes of IT support and removes the single point of failure entirely.

All three divisions are rated Medium. The rating applies specifically to the MHU tracking sheet as a data product, not to each division’s wider clinical or regulatory systems. PHS records every patient encounter in Athena, a contracted EMR system not tied to any individual staff account. EH records every blood lead test result in EpiTrax, a state-managed mandatory reporting system. CH’s Diabetes Prevention Programme data flows to the CDC under federal reporting requirements. Each of these systems provides strong, independent protection for its own clinical and regulatory data. However, none of them protect the tracking sheet data, which is what this dimension is rating. The aggregate event-level programme data in the shared Excel workbook is exposed to exactly the same OneDrive single-point-of-failure risk for all three divisions equally, and the rating reflects that shared vulnerability.

Joint Display: *Convergence*

Table 37: Joint Display 5: The OneDrive Single-Point-of-Failure Risk

Strand	Evidence	What it establishes
Qualitative	EH written response confirms data loss has already happened once when a staff member's OneDrive account was deactivated after leaving the county. A separate follow-up with the Data Team indicates ownership has since moved to a different individual, with final confirmation still pending.	The risk is not theoretical; it has a documented prior occurrence within this programme's own history, and the fact that ownership has already transferred once shows the structural risk persists regardless of who currently holds the file.
Quantitative	SAS PROC CONTENTS (pp. 141–143): the dataset contains no version history field, no backup timestamp field, and no owner-of-record field among its 49 variables.	The tracking sheet's own structure confirms there is no quantitative safeguard against the qualitative risk: nothing in the data architecture would catch or flag a future loss event before it happened.

Integration outcome: convergence. The two strands agree from different directions: the qualitative account confirms the risk is real and has already materialised, and the quantitative inventory confirms no structural safeguard exists to prevent recurrence. Recommendation R11 follows directly from this joint finding.

Table 38: Dataset Inventory (SAS PROC CONTENTS, pages 141–143).

Property	Value	Confirmed from SAS
Dataset name	WORK.MHU	SAS PROC CONTENTS, p.141
Observations	85	Confirmed
Variables	49	Full list, pp.142–143
Created	06/25/2026	
PII variables	0 of 49	All 49 variable names reviewed
Data type	Event-level aggregate	No individual patient records

5.11 Confidentiality – All Divisions: HIGH

The master tracking sheet contains no patient names, dates of birth, addresses, or individual identifiers in either workbook. All 49 variable names were reviewed against SAS PROC CONTENTS (pages 141–143): none are personally identifiable. Individual records are held in regulated systems. Across all three divisions, staff were consistent and specific about how individual data is kept separate from the aggregate tracking sheet.

Public Health Services: “ We will get you de-identified data – we are not going to pull anything identifying. ”

Community Health: “ We do not keep their individual health information. That is also not something the community health department wants to keep. ”

Any demographic data pulled from Athena for Phase 3 will be de-identified before transfer under the HIPAA data use agreement being coordinated as of June 2026.

6. Cross-Cutting Findings

The staff hours documentation conflict is the most important finding from reading both workbooks in full. The main Read Me and the PHS/CH tabs give opposite instructions for the same variable. Resolving this requires one decision from leadership, followed by updating all three Read Me tabs to say the same thing.

The Read Me documents are a genuine programme strength. Comprehensive written definitions for all key variables, established through group discussion before operations started, is uncommon in first-year programmes. This foundation can be maintained and extended.

The EpiTrax discovery is the most promising finding in the assessment. EpiTrax holds approximately 150 data points per child for every blood lead test in Weld County. EH staff offered to pull all MHU-related records. This is the first priority after this report is submitted.

Services invisible in the notes field. Narcan, fentanyl test strips, condoms, PrEP navigation, and pregnancy tests appear only in free-text notes. These real services are invisible to any structured analysis or dashboard.

CH engagements: not a decline. The raw total fell from 692 to 439 (37%). Annualised (1,038 vs 1,054) the programme is flat. Per event, CH delivered more in 2026 (median 31 vs 14). This must not be presented as a performance decline.

Joint Display: *Convergence via correction*

Table 39: Joint Display 3: CH Engagement, Apparent Decline Corrected

Strand	Evidence	What it establishes
Quantitative (naive)	Raw totals: 692 (2025) vs 439 (2026), a 37% drop.	Read alone, this looks like a real decline in CH performance.
Quantitative (corrected)	SAS PROC PRINT (p. 58): annualised 1,038 vs 1,054. Per-event median rose from 14 to 31 (SAS PROC NPAR1WAY, p. 120, $p = 0.079$, $r = 0.345$).	The decline is an artefact of comparing eight months of 2025 to five months of 2026; per-event delivery actually rose.
Qualitative	CH staff attended 29 events in 2025 versus only 13 in 2026, a function of the shorter 2026 observation window and event scheduling, not a qualitatively reported drop in CH effort or capacity.	No interview evidence supports a narrative of CH withdrawal or reduced effort; the qualitative record is silent where it would have spoken up had something changed operationally.
Integration outcome: convergence via correction. The naive quantitative read and an unprompted qualitative narrative would have agreed on a false story (CH in decline). Re-running the quantitative analysis correctly (annualised, per-event) and checking it against the qualitative record together rule that story out.		

The North region disappeared in 2026. Five events were held in North in 2025. Zero in the first five months of 2026. No explanation appears in the data or interviews.

CHW contributions are invisible. Community Health Worker hours are tracked in a separate document and do not appear in the master tracking sheet.

The OneDrive ownership risk. The file has already changed individual ownership once, from the original PHS staff member who left the county to a different individual in a different division, with confirmation of the current owner still pending. A county-wide transition to SharePoint is planned, which would permanently resolve this issue by removing individual file ownership entirely from the equation. That transition was originally scheduled for 2026 but has been delayed because IT is currently managing the county-wide rollout of Microsoft Copilot AI. Until SharePoint is available, transferring the master file to a shared Data Team account is the immediate fix, takes approximately 30 minutes with IT support, and eliminates the single-point-of-failure risk regardless of who currently holds it.

The two-sheet PHS system has never been audited. PHS maintains two separate tracking sheets, a general outreach sheet and the MHU master sheet, with events transferred manually between them. If an MHU event is sitting in the outreach sheet but was never transferred to the master sheet, the programme dataset is silently missing that event, and no process currently exists to check whether this has happened. Recommendation R15 addresses this directly.

7. Recommendations

7.1 Impact Goal 1: Know Who You Serve

- R1.** Request the EpiTrax demographic data pull from EH for all MHU blood lead test records from April 2025 to the present.
- R2.** Contact Olivia Egen or Kelly Martinez for a CDC referral for a formal DPP participant data request.
- R3.** Arrange a de-identified Athena demographic pull for all patients seen at MHU service locations.
- R4.** Add an Event Population Category field to the tracking sheet.

7.2 Impact Goal 2: Trust What You Measure

- R5.** Resolve the staff hours documentation conflict by updating all three Read Me tabs.
- R6.** Convene a group meeting to agree the engagement definition in practice and update the Read Me.
- R7.** Update the CH health screenings Read Me to list all eight tests currently offered.
- R8.** Create an EH Read Me tab using the same structure as the PHS and CH tabs.
- R9.** Add people turned away and services delivered as structured columns in the PHS section.
- R10.** Interview frontline staff directly about the address intake process at events (form type, who collects it, whether the field is treated as optional under time pressure) to confirm or rule out the staff education explanation offered in the project inception meeting for the 2025–2026 decline in address completeness documented in Section 5.7.2.

7.3 Impact Goal 3: Build Systems That Last

- R11.** Transfer the master tracking sheet from the individual staff OneDrive account to a shared Data Team account.
- R12.** Assign a unique sequential Event ID to every event, retroactively and going forward.
- R13.** Establish a schema change protocol requiring Data Team approval before any column is added, removed, or renamed.
- R14.** Distribute the Read Me documents at every quarterly mobile unit planning meeting as a standing agenda item.
- R15.** Cross-reference the PHS outreach tracking sheet against the MHU master sheet.
- R16.** Follow up with Community Health on the CH engagement finding before presenting it to leadership.

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